Executive Commercialization Assessment Report

A physics-constrained Bayesian inversion framework for ion energy flux at a plasma sheath

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Prepared for an incubation and investment committee, and for domain reviewers.

Subject technology	Multi-diagnostic Bayesian inverse framework for plasma–surface ion energy flux
Stage	Post-benchmark, pre-laboratory-validation [C]
Assessed readiness	TRL 2, on the team's reported execution of the closed-loop synthetic benchmark [C]. See §9.2.
Revenue	None. Pre-revenue, pre-laboratory-validation.

How to read this document. Every sentence containing a number ends in one of four tags. [C] marks a claim made by the technology team and *not independently verified*. [CALC] marks the analyst's own arithmetic, with its inputs and formula shown, and [A] marks the analyst's own retrieval, enumeration or search result, a count or an absence established by this assessment rather than quoted from a source. A bracketed numeral such as [S1] marks an external source, listed in full in Appendix E with a live URL and the date it was opened. Sentences beginning **Assessment:** are the analyst's judgement, not fact. Sentences beginning **Projection (scenario):** are forward-looking and state their assumption in the same sentence. **No number appears anywhere in this document without one of these tags.**

Quotation policy. Material inside quotation marks was read at source and is reproduced verbatim. Where a source is characterised without quotation marks, the wording is this report's own paraphrase. Where a specific claim could not be verified at source, that is stated at the claim itself.

The central caution. The team reports the closed-loop synthetic benchmark executed, and has supplied its results [C]. **Every one of those figures is synthetic, self-reported and not independently verified**, and no measurement against real laboratory data exists. The compute projection remains a projection. Any reading of this document that treats a synthetic benchmark result as a laboratory result, or a projected number as an achieved one, is a misreading.

Executive Summary

The problem is real, quantified, and currently unsolved. Ion energy flux at a plasma–surface boundary governs erosion, heating and lifetime in electric thrusters, semiconductor etch tools and fusion plasma-facing components, and it cannot be measured without disturbing it. A Langmuir probe in a low-pressure argon plasma is expected to underestimate plasma density by about 50 % under stated reference conditions [S1], and a 2020 community review states plainly that plasma thruster development "rely on a semi-empirical approach, combined with long and expensive lifetime tests" [S2]. The cost of that gap is measurable in calendar years: NASA's NEXT ion thruster life qualification ran from June 2005 to September 2013, accumulating 50,170 h of operation and 902 kg of xenon before being voluntarily terminated under budget constraints, having met or exceeded its original goals [S3].

The differentiator survives verification, and it sits in a more precise place than the team's framing. Bayesian fusion of multiple plasma diagnostics is not novel: it was published for magnetic confinement fusion in the peer-reviewed literature by 2010 [S4], and the Minerva framework at Wendelstein 7-X fuses the same number of channels as this project proposes [S5]. What none of the frameworks read for this report does is infer ion energy flux *at a surface* through a sheath forward model. That single element, not the Bayesian method and not the channel count, is the defensible position.

The commercial case rests on counted customers, not on a market report. No credible published figure for a "plasma diagnostics market" exists; the eight major research houses named in §5.1 were searched and none publishes one **[A]**. The serviceable market was therefore built bottom-up from named, enumerated organisations and comes to 84 [CALC, §5.3], counted from a dated archival snapshot so that the figure can be re-checked. At the one published comparable seat rate this yields between £254,100 and £762,300 per year [CALC, §5.3]. Licence revenue alone does not build this company, and the non-dilutive instruments this applicant can actually use are domestic Indian schemes, not United States SBIR, which is unavailable to it (§15.2).

The dominant risks are not technical failure but novelty and access. The patent record in this niche is thin, but that is a novelty *risk* rather than open whitespace: the corresponding non-patent literature is dense, and dense non-patent prior art is a validity problem (§11.2). Separately, no plasma diagnostics group was found on any of the Indian institutional pages checked other than one (§12), making access to a validation facility a live execution risk.

The gate this assessment named has been cleared. The project is at **TRL 2**: all four clauses of NASA's software criterion are now reported met, including the two evidentiary ones, working code and experiments on synthetic data (§9.2) **[C]**. That is the milestone this assessment identified as the one that would change its status, and it has been reached. The team reports the honest T2 tier at **3.99 % ± 2.76 % mean relative error on ion energy flux** over 10 noise realisations, with **8 of 10 credible intervals covering at the 90 % level** **[C]**. That sits inside the 6 %/5 %/30 % triple the nearest published comparable reports [S6]. **Assessment:** On the team's own figures the calibrated-uncertainty claim survives its first real test, and the recommendation moves from *run the benchmark* to *close the identifiability gap and get this independently reproduced*.

The finding that is worth more than the headline number. The team reports that naive four-channel fusion is *worse* than three channels: 25.7 % error at 0 of 10 coverage, with credible intervals four times narrower than the configuration that gets the right answer, i.e. confidently wrong **[C]**. A per-channel audit isolated one channel, OES, as understating its own uncertainty by a factor of 29 to 138 **[C]**. **Assessment:** Two things follow, and they point in opposite directions. The unfavourable one: the report's own differentiator was stated as four fused channels, and channel count alone is now shown by the team's own data not to be the thing that works. The favourable one, which is larger: **down-weighting rather than dropping the channel recovers 3.99 % at 8 of 10 coverage with all four retained** **[C]**, and a team that publishes the configuration that made its own headline claim fail is displaying the behaviour this class of investment most needs and least often gets.

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What is genuinely strong here. The team's own posture is the most creditworthy thing in the file. It pre-commits to reporting the pessimistic tier only, designs deliberate model mismatch in to avoid inverse crime, and gates its own compute projection for day-one correction **[C]**. It then honoured all three: T2 is the tier it quotes, it reports the negative four-channel result against its own stated differentiator, and it records the identifiability map as not computed rather than implying otherwise **[C]**. Its self-assessment of "TRL 1–2", made before the benchmark ran, was defensible against the official wording rather than inflated (§9.2), which against the norm at this stage counts in its favour. Documentation maturity materially exceeds every other axis (Figure 9.1), unusual, and in the right direction.

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1 Problem Definition

1.1 The measurement that cannot be made

Ion energy flux across a plasma sheath sets the rate at which a surface is heated and eroded. In an electric thruster it determines channel wall wear and therefore mission life; in a plasma etch tool it governs feature profile and selectivity; in a fusion device it drives divertor and first-wall erosion. The quantity is causal for all three industries, and in none of them can it be measured directly without perturbing it.

Two measurement routes exist, and both fail in a specific, documented way.

1.1.1 Intrusive probes perturb the quantity they measure

The headline number for this dossier's problem statement: a Langmuir probe measuring a low-pressure argon plasma at approximately 100 mTorr, with probe radius 0.1 mm, $kT_e = 3$ eV and $n = 3 \times 10^{10}/\mathrm{cm}^3$, is expected to **underestimate plasma density by about 50 %** [S1].

The authors of that result argue it "may resolve the long-standing problem of why plasma densities obtained with Langmuir probes are consistently lower than those obtained in microwave diagnostics" [S1]. A companion study reports that plasma density at the probe surface fell to about one fifth of the unperturbed value, with measured electron temperature about 40 % below the unperturbed plasma [S7]. Restating the first of those as a depletion gives 80 % [CALC: $1 - 1/5 = 0.80$]. **Assessment:** Two caveats the source itself supplies: [S7] reports these values citing earlier work rather than measuring them, and describes perturbations of that size as typical of *collisional* plasma, a different regime from the low-pressure case that supplies this chapter's headline number.

The problem compounds for derived quantities. Recovering an electron energy probability function to better than 3 % error requires the circuit resistance ratio $R_c/R_{p,\mathrm{min}} < 0.01$ [S8]. **Assessment:** That source is a preconference tutorial delivered at the Gaseous Electronics Conference, Princeton, 28–29 September 2013 [S8]; it retrieves normally over HTTPS and the figure was re-verified in it directly. The tutorial itself points to the peer-reviewed carrier for these results [S9], which is cited here alongside it. The same tutorial states that "there is no plasma diagnostics method other than probe diagnostics where the danger of incorrect measurements and erroneous interpretation of results are so great" [S8].

Assessment: The severity is best evidenced not by any single error bar but by the fact that the electric propulsion community found it necessary to publish three separate formal "Recommended Practice" papers for probe use (for Langmuir, emissive and Faraday probes respectively) in one 2017 issue of a single journal [S10]. Fields do not codify practice that is working reliably.

For the sheath specifically, probes disagree with each other. Plasma potential measured by cylindrical and planar Langmuir probes differs from that measured by emissive probes in the neighbourhood of the presheath near a negatively biased electrode, and emissive probes are described as "the only known electrostatic probe technique able to measure the plasma potential accurately throughout the presheath" [S11].

[DATA GAP 1] *claim needed:* The magnitude, in volts, of the documented Langmuir-versus-emissive probe disagreement inside the presheath. *Why unavailable:* The abstract confirms the disagreement qualitatively and states it is not proportional to T_e, but the numerical magnitude sits behind the publisher's paywall. *how to get it:* Institutional library access to Plasma Sources Science and Technology 29(2) 025015, or direct correspondence with the authors

1.1.2 Individual optical diagnostics see only part of the state

Non-intrusive optical diagnostics do not perturb the plasma, but each returns a partial observable. A 2020 review states that for compact, energetic Hall thrusters "probes can strongly perturb the plasma and it

becomes difficult to measure plasma parameters in the ionization and acceleration zones", and that optical, non-invasive diagnostics "are required to investigate the plasma-wall interaction effects in these high-density and high-energy plasmas where probes hardly survive. This is difficult and is a major challenge for the community" [S2].

Assessment: Optical channels also disagree with probes. The literature reports that in Hall thruster plumes the laser-induced fluorescence signal can collapse while ion density at the same location remains finite, attributed to the strong electron-temperature dependence of the metastable excitation cross-section. **This report does not cite a source for that statement**, because the paper was not opened at source; it is recorded here as context and carries no number.

Assessment: The problem is therefore not that either route is useless. It is that the intrusive route corrupts the measurement and the non-intrusive route underdetermines it. That structure (partial, non-perturbing observables plus a known physical model) is the textbook setting for an inverse problem, which is what makes the proposed approach appropriate in principle regardless of whether this particular team executes it well.

1.2 What the gap currently costs

1.2.1 Electric propulsion: qualification measured in calendar years

NASA's NEXT Long-Duration Test ran from **June 2005 to September 2013**. At termination the thruster had accumulated **50,170 h** of high-voltage operation, processed **902 kg** of xenon and delivered **34.9 MN s** of total impulse [S3]. That is an elapsed duration of about **8 years 3 months** [CALC: June 2005 to September 2013] for a single thruster life demonstration.

The test "exceeded its original qualification throughput requirement of 450 kg in December 2009", and termination began in April 2013 "in light of recent budget constraints" [S3]. Required operational lifetimes for ion thrusters are stated as "on the order of 10,000 to 100,000 h", with the NEXT qualification requirement of 450 kg corresponding to roughly 22,000 h of operation at the highest throttling point [S12].

The successor 12.5 kW HERMeS Hall thruster is designed for an operational lifetime "exceeding 50 kh" [S13]. Its Long Duration Wear Test accumulated approximately 3570 h between October 2017 and October 2018, and its stated goal was "to identify and correct design or facility issues prior to the flight qualification campaign" [S14]. Two earlier wear tests of approximately 1700 h (2016) and a series of seven 200 h segments (2017) preceded it [S14].

Assessment: One multi-thousand-hour wear test and two shorter campaigns (approximately 3570 h, 1700 h and seven segments of about 200 h **[CALC]**) were run purely as a pathfinder *before* the qualification campaign began. The cost being carried is not the price of a diagnostic; it is the schedule cost of resolving by endurance testing what cannot be resolved by measurement.

[DATA GAP 2] *claim needed:* The dollar cost of a single electric propulsion life-qualification campaign. *Why unavailable:* No dollar figure appears in any of the NASA NEXT or HERMeS documents reviewed; the literature is uniformly qualitative on cost, using phrases such as "long and expensive lifetime tests". *how to get it:* Query USAspending.gov for the Aerojet Rocketdyne AEPS contract award value, which is the closest available public proxy; or request the figure directly from NASA Glenn

Erosion sensitivity is itself steep, which is what makes the prediction problem hard: erosion of the HERMeS inner and outer front pole covers varies "by 76 to 300 percent as a function of discharge voltage and by up to 40 percent as a function of magnetic field strength", and shifting cathode position upstream reduced keeper erosion rates by 84 % [S13].

1.2.2 Fusion: heat flux limits set by an unmeasurable quantity

The ITER divertor vertical targets are designed to sustain 10 MW/m² in steady state and 20 MW/m² during slow transients, a heat load the ITER Organization describes as "ten times higher than that of a spacecraft re-entering Earth's atmosphere" [S15]. The divertor comprises 54 cassette assemblies of about ten tonnes each, to be "replaced at least once during the machine's lifetime" [S15].

1.2.3 Semiconductor: the cost figure everyone quotes and nobody sources

[DATA GAP 3] *claim needed:* The cost of semiconductor fab downtime per hour, and the cost of plasma process excursions. *Why unavailable:* Every per-hour figure encountered in research (ranging from $2 million to over $8 million per hour) traced back to search engine optimisation content farms and vendor marketing blogs with no primary citation. No figure was found in any SEMI publication, IEEE Transactions on Semiconductor Manufacturing paper, or named fab disclosure. *how to get it:* A SEMI or SEMATECH publication, a peer-reviewed IEEE Transactions on Semiconductor Manufacturing paper, or a named fab's own public disclosure. This is the most commercially significant open gap in the dossier

[DATA GAP 4] *claim needed:* The average selling price of a plasma etch tool. *Why unavailable:* Lam Research, Applied Materials and Tokyo Electron do not publish per-tool prices; the figure circulating for one recent tool appears only in a paywalled analyst newsletter's search snippet and could not be read in the article itself. *how to get it:* Analyst subscription (SemiAnalysis, TechInsights), or a fab equipment procurement record obtained under a freedom-of-information request

Assessment: A peer-reviewed figure for chamber-to-chamber etch rate variability was located but **deliberately excluded from this report**. It was surfaced from search-engine text of a paywalled paper rather than read in the article, and this dossier's own evidence rules do not permit quoting a number that was never read at source. The two data gaps above therefore carry this section on their own, and the semiconductor value argument rests on no quantified cost claim at all.

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2 Value Proposition

2.1 The proposition, stated against a named baseline

The baseline is not "nothing". It is a retarding field energy analyser inserted into the plasma. Against that baseline the proposition is:

1. **It is designed to leave the plasma undisturbed.** All four channels are specified as non-intrusive [C], though two of them (Thomson scattering and laser-induced fluorescence) inject laser energy, and **no perturbation measurement exists** because nothing has been run. The baseline's perturbation is *expected*, on a first-principles model under stated reference conditions, to reach about 50 % density underestimation [S1], an analytic expectation, not a measurement.
2. **It returns a calibrated posterior, not a point estimate.** **Assessment:** On the evidence checked, a calibrated interval is an unadvertised attribute in the commercial field: no accuracy figure or uncertainty budget for the ion energy quantity was found on any vendor page checked for this report (§7.2.2). The stronger claim that it is absent from the field would require auditing vendors not checked here.
3. **It states what it cannot resolve.** Identifiability is reported via Fisher information and null-space analysis rather than assumed [C].
4. **It reaches where hardware cannot go.** Optical access is possible in geometries where "probes hardly survive" [S2].

Assessment: Item 3 is the most commercially underrated and the most defensible. A framework that declares its own blind spots is more useful to a qualification engineer than one that returns a confident number everywhere, because qualification decisions are margin decisions. It is also the hardest claim for a competitor to retrofit.

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2.2 What the proposition is not

Assessment: Three claims that might appear natural are *not* supported by the evidence gathered, and should not be made:

- **Not "the first Bayesian fusion of plasma diagnostics".** Integrated Data Analysis has been applied at ASDEX Upgrade since 2004 [S16]. **Assessment:** That date is taken from a background slide in a tutorial lecture rather than from a peer-reviewed methods paper, so it should be read as indicative of when the work began rather than as a precise institutional milestone; the peer-reviewed record of the method itself is [S4] and [S17].
- **Not "more diagnostic channels than anyone else".** Minerva at W7-X fuses four channels [S5], the same count this project proposes [C]; Minerva at JET models about ten diagnostic systems [S18]. **Assessment:** The team's own benchmark has since corroborated this from the inside: adding the fourth channel naively made the result worse, not better (§8.2) [C]. Channel count was already the wrong place to look for the differentiator, and the project's own data now says so.
- **Not "faster".** Neural network surrogates of Bayesian diagnostic models are already published for this exact problem class [S19], and a Hall thruster group achieved roughly a tenfold runtime reduction through a combination of internal code optimisations and coarsened grid resolution, making direct Bayesian inference practical without a surrogate [S20].

The claim that survives is precise and specific: *inference of ion energy flux at a surface, through a sheath forward model, from non-intrusive optical channels only*. Chapter 7 verifies that this position is unoccupied by every retrieved framework.

3 Technology Positioning

3.1 Category definition

Assessment: The technology does not fit an existing analyst category, and this is a structural fact rather than a presentational problem (§5.1). The closest accurate description is *model-based inferential instrumentation*: software that converts an incomplete set of physical measurements into an estimate of an unmeasurable quantity, with a stated uncertainty, by inverting a first-principles forward model.

3.2 Architecture as stated by the team

The following is the team's description and has not been independently verified or executed **[C]**:

- Observation model $y = F(x) + \varepsilon$, with F built as a strict four-layer chain: plasma physics, then emission and scattering, then optical transport, then detector **[C]**.
- Inversion via $p(x|y) \propto p(y|x)\, p(x)$, with sheath theory entering as a *structural* prior (the sampler can only propose states satisfying Poisson's equation) rather than as a soft penalty term **[C]**.
- Six inference engines (MAP, Laplace, NUTS, SMC, EnKF, particle filter) behind one interface, for cross-validation **[C]**.
- Four fused optical channels: optical emission spectroscopy, laser-induced fluorescence, Thomson scattering, interferometry **[C]**.

Assessment: The structural-prior choice is the technically distinctive decision. Imposing Poisson's equation as a constraint on the proposal space rather than as a likelihood penalty is a stronger guarantee: it makes physically impossible states unreachable rather than merely improbable. Whether it is implementable efficiently at the stated scale is unknown, because nothing has been run.

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3.3 Design-level figures the team reports

These are closed-form design calculations, not measurements or results **[C]**:

- Thomson scattering photon budget of 0.16 photoelectrons per shot at $n_e = 1 \times 10^{17}/\mathrm{m}^3$ at the stated reference point **[C]**.
- Interferometry detection floor of $8.4 \times 10^{15}/\mathrm{m}^3$ over a 400 mm chord **[C]**.
- Per-channel calibration uncertainties of 6 % to 7 % **[C]**.

The one number that must not be misread. The team reports a full simulation programme (an approximately 5,000-point training ensemble plus benchmark, ablation and coverage suites) projected at about 33 h wall-clock on a single NVIDIA RTX A4000 GPU **[C]**. This rests on an *assumed, not measured*, GPU particle-push throughput. Gate G-1.4 has since been reported met for the L1 fluid solver, at 86 ms per solve against an estimate of about 1 s, but **the L2 particle-push throughput on which the 33 h figure actually depends was not measured**, and the team carries it forward to G-2 rather than closing it **[C]**. It remains a projection. It is not a runtime.

Assessment: That the team flags this itself, rather than being caught at it, is among the strongest credibility signals in the submission. The instinct to gate one's own weakest assumption is uncommon at this stage.

4 Industry Mapping

Each sector below is included only where a cited fact establishes that the quantity in question actually matters to it.

Industry	Cited justification that sheath ion energy flux matters	Evidence
Electric propulsion	"Sheaths near the wall determine ion energy of ions impinging the walls and therefore, the sheath structure and sheath potential affect the wall erosion and, correspondingly the thruster lifetime."	[S2]
Fusion plasma-facing components	Divertor targets designed for 10 MW/m² steady state and 20 MW/m² in slow transients; 54 cassettes to be replaced at least once in machine life.	[S15]
Semiconductor plasma etch	Applied Materials holds a granted patent on *creating* multi-peak ion energy distribution functions by pulse-burst modulation; the industry actuates on this quantity, so it has commercial reason to measure it.	[S21]
Plasma equipment manufacture	Lam Research holds a granted patent whose claim 1 determines sheath capacitance from a total-charge value "determined based on an optical signal sensed by an optical sensor", and drives an RF generator until that capacitance meets a target [S22]. The industry already closes a control loop on an optically derived sheath quantity.	[S22]
Space situational / flight instrumentation	NASA sought market research on a plasma diagnostic package for the Gateway Power and Propulsion Element, to measure "ion and electron kinetic energies, ion and electron fluxes, rates of surface erosion and deposition".	[S23]

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Assessment: Sectors deliberately excluded, despite being plausible-sounding: plasma medicine, agriculture and water treatment. No cited evidence was found that sheath ion energy flux is a governing design quantity in any of them, and listing them would be sector-name padding rather than industry mapping.

5 Market Research

5.1 The category does not exist

The single most important market finding. There is no credible published "plasma diagnostics market size". All eight major market-research houses were searched (Grand View, MarketsandMarkets, Precedence, Global Market Insights, Mordor, Fortune Business Insights, Allied Market Research and IMARC) and **none publishes such a report**. The only sources offering one are automated report mills whose figures for nominally the same category differ by roughly a factor of seven, one such pair quotes 67.8 million US dollars and 485 million US dollars for nominally the same category in adjacent base years **[CALC]**. Those mills are not cited anywhere in this report.

Assessment: This cuts both ways and the committee should weigh both directions. Negatively: there is no analyst tailwind, no category budget line that a customer already owns, and no third-party number a founder can point at. Positively: no incumbent has been sized, funded and positioned against this category either. On balance it is a go-to-market risk, and it is the reason this chapter builds from counted units rather than from a top-down figure.

5.2 Adjacent markets, as context only

The figures below size the surrounding industries. **None of them is addressable by this product**, and quoting any of them as a total addressable market would be the substitution this report exists to avoid.

- Wafer fab equipment: 115.7 billion US dollars in 2025, forecast to 135.2 billion in 2027 [S24].
- Total semiconductor equipment: 133 billion US dollars in 2025, forecast to 156 billion in 2027 [S24].
- Fusion industry funding: 4.48 billion US dollars raised in the twelve months to July 2026, taking total funding to date to 14.24 billion [S25]. **Assessment: That total is not all private money.** The same page splits it explicitly: **13.26 billion private and 0.98 billion public** [S25]. "To date" rather than "since 2021": the source's own chart shows the total includes pre-2021 investment.
- Space propulsion market: 10.21 billion US dollars in 2024, forecast to 20.02 billion by 2030 at 11.9 % compound annual growth [S26].
- Satellite propulsion market: 2.60 billion US dollars in 2024, forecast to 5.19 billion by 2030 [S27].

Assessment: Note that no fusion *revenue* figure is quoted, because there is no commercial fusion electricity revenue to quote. Investment inflow is the honest metric for that sector, and reports forecasting fusion energy market revenue to 2046 are forecasting a product that does not exist.

5.2.1 Two published market figures this report refuses to use

Assessment: Two figures encountered during research are documented here as rejected, because a reader may find them independently and wonder why they are absent.

1. Mordor Intelligence reports an electric-propulsion-satellite market of 13.03 billion US dollars for 2025 [S28]. That figure is *byte-identical* to the same house's figure for the entirely unrelated semiconductor metrology and inspection market in the same base year, also 13.03 billion US dollars [S29]. Both pages were opened independently to confirm the coincidence. **Assessment:** It is also implausible on its face, since it would make electric propulsion satellites roughly five times larger than the entire satellite propulsion market of 2.60 billion US dollars [S27] **[CALC]**.
2. Fortune Business Insights' electric-propulsion-satellite series does not reconcile with itself: the same page states a 2025 base of 1.86 billion US dollars, then a 2026 figure of 1.28 billion (roughly 31 % lower

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[CALC]) then a 2034 forecast of 3.09 billion at a stated 6.56 % compound growth rate, a path that in fact requires closer to 11.7 % **[CALC]** [S30].

5.3 Serviceable market, counted from named organisations

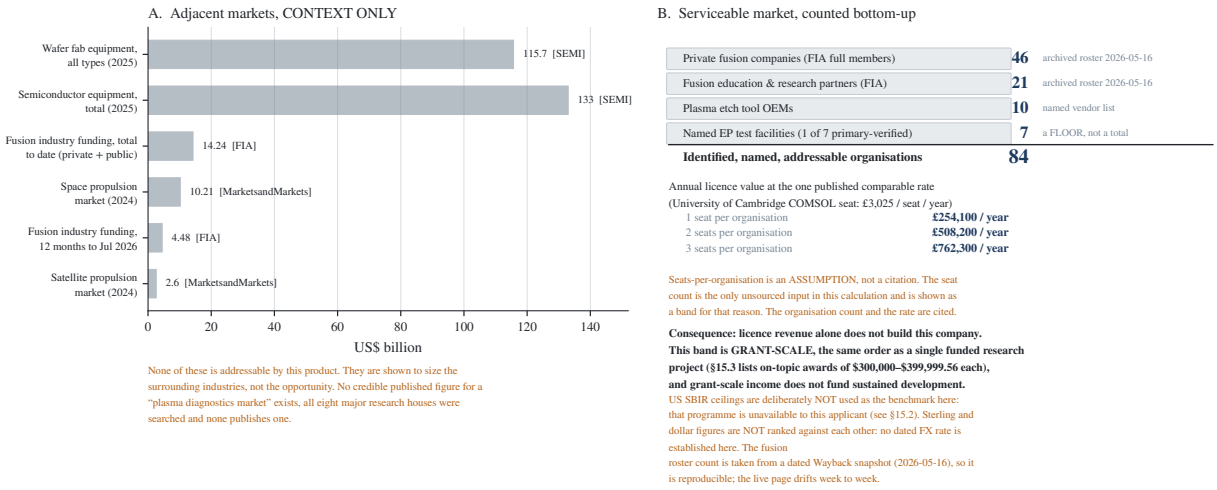


Figure 5.1: Market sizing. Panel A shows adjacent industries as non-addressable context; Panel B builds the serviceable market from enumerated organisations and shows the licence arithmetic that follows. Source: [S24], [S25], [S26], [S27], [S31], [S32], and **[CALC]**. Underlying data: Appendix F, Table F.3.

The addressable organisations were counted, not estimated:

- 46 private fusion companies, enumerated from a **dated archival snapshot** of the Fusion Industry Association full-member roster taken on 16 May 2026 [S33].
- 21 fusion education and research partners, from the same snapshot [S33].
- 10 plasma etch tool manufacturers **[A]**, enumerated by name: Lam Research, Applied Materials, Tokyo Electron, Hitachi High-Tech, SPTS/KLA, Oxford Instruments, Plasma-Therm, Samco, AMEC and NAURA.
- 7 electric propulsion test facilities **[A]**, named individually: NASA Glenn Vacuum Facility 5, NASA JPL Owens, Georgia Tech HPEPL, University of Michigan PEPL, The Aerospace Corporation, QinetiQ LEEP and the Colorado State University Orion chamber. **Assessment: Only one of the seven (Georgia Tech HPEPL [S34]) was confirmed by opening its own facility page.** The Michigan PEPL page rejected automated access (DATA GAP 12, §14), and the remaining five were identified from secondary references rather than from primary facility pages. **This is a floor, not a total, and it is the least well-evidenced component of the 84.**

Serviceable market = 46 + 21 + 10 + 7 = 84 **identified, named organisations [CALC].**

Assessment: This number is reproducible. The membership roster supplying 67 of the 84 is a live page that drifts week to week, so the counts above are taken from a **dated Wayback Machine snapshot of 16 May 2026** [S33] rather than from the live page, anyone can re-open that snapshot and recount. **Assessment:** Two judgements are declared rather than buried: the roster's affiliate-member category (110 entries in the same snapshot) is *excluded*, because affiliates are suppliers, utilities and service providers rather than operators of plasma devices; and members were counted by distinct logo assets, which mis-counts any member listed without a logo or with more than one.

[DATA GAP 5] *claim needed:* An enumerated list of the 46 member organisations by name. *Why unavailable:* The count is now anchored to a dated archival snapshot and is reproducible [S33], but members were counted by distinct logo assets, which mis-counts any member listed without a logo or with more than one. *how to get it:* Transcribe the member names from the archived snapshot into an appendix, converting a logo-asset count into a name list

[DATA GAP 6] *claim needed:* Awarded federal contract values for every named incumbent diagnostics vendor, not only Impedans. *Why unavailable:* Only one vendor's award history was queried; the equivalent query was not run for Hiden Analytical or the other named suppliers, so the displaced-spend cross-check rests on a single recipient. *how to get it:* Re-run the same awards API query, filtered by recipient name, for each named incumbent, and report the full range rather than one vendor's

[DATA GAP 7] *claim needed:* A worldwide census of electric propulsion and thruster vacuum test facilities. *Why unavailable:* No authoritative count exists in any public source searched; named facilities were verified individually but no register or survey was found. *how to get it:* A manual review of International Electric Propulsion Conference proceedings, which is the likeliest venue for such a survey, or a direct enquiry to the Electric Rocket Propulsion Society

5.3.1 Price anchor, and the licence arithmetic

The only published, auditable, institutional price found for comparable scientific software is the University of Cambridge Department of Engineering's published COMSOL rate: £3025 per seat per year, on a 30-seat, three-year contract running 1 March 2026 to 28 February 2029 [S32].

Projection (scenario): Assuming one seat per organisation, annual licence value is 84 × *$*3,025 = *$*254,100; at two seats, £508,200; at three seats, £762,300 **[CALC]**. All three figures are in pounds sterling, matching the currency of the only published comparable rate [S32].

Assessment: Seats-per-organisation is the only unsourced input in that calculation, which is why it is presented as a band rather than a point. The organisation count and the seat rate are both cited. The band should be read as an *upper* bound in any case, since it assumes universal adoption across every identified organisation.

The consequence, stated plainly. Licence revenue at these volumes is *of the same order as a single research grant, not a multiple of it*: the band runs £254,100 to £762,300 per year **[CALC]**, against on-topic awards of $300,000 to $399,999.56 each [S35], [S36], [S37]. **Assessment:** The two figures are in different currencies and this report does not establish a dated exchange rate, so they are deliberately not subtracted or ranked, see the data gap below. The conclusion that survives without any conversion is the one that matters: licence income on this customer base is grant-scale, and grant-scale income does not fund sustained software development. Note also that the United States SBIR ceilings often used as this benchmark are *not* available to this applicant (§15.2).

[DATA GAP 8] *claim needed:* A dated GBP/USD exchange rate, to permit direct comparison of the sterling licence band against dollar-denominated research awards. *Why unavailable:* No exchange rate was retrieved or cited during this assessment, and using an undated rate would introduce an unsourced number into the report's central commercial comparison. *how to get it:* Take a published central-bank or exchange reference rate for a stated date and restate both figures in one currency; note that against the largest on-topic award of $399,999.56 the break-even is approximately 0.52 USD per GBP **[CALC]**, so at every realistic rate the top of the licence band exceeds that award; the ordering is not rate-dependent, and the defensible statement is that the two are of the same order of magnitude

5.4 Cross-check: is the bottom-up number consistent with anything?

Assessment: A top-down number cannot be constructed for comparison, because the category does not exist (§5.1). The nearest available consistency check is on the *displaced spend* rather than the market. Seven awarded United States federal contracts **to Impedans Ltd** range from $16,240 to $86,080, clustering at $30,000 to 40,000 for a single instrument; these were retrieved by querying the awards API for that named recipient rather than by browsing the site [S38]. A distinct vendor publishes a full public price list on which a retarding potential analyser (the direct instrument analogue) costs $2950 for the probe alone and $4450 with feedthrough and cables [S39].

Assessment: These two anchors agree with each other *for the vendors they cover*: a single Impedans instrument is a low-thousands-to-low-tens-of-thousands purchase rather than a capital-equipment purchase, and the one published price list in the industry sits in the same band. **They do not characterise the incumbent spend as a whole.** The equivalent query was not run for the other named incumbent, Hiden

Analytical, whose systems are materially more complex (an energy analyser combined with a quadrupole mass spectrometer) and would be expected to command higher awards. The cross-check therefore establishes a floor for one vendor, not a ceiling for the category, and it should not be read as proof that the displaced spend is small.

TEAM ASTROPHAGE

6 Existing Solution Analysis

6.1 Commercial instruments

Only two product families in the retrieved evidence measure the target quantity, and both do so by inserting hardware into the plasma.

Product	What it does	Published specification
Impedans Semion retarding field energy analyser	Measures ion flux and ion energy distribution striking a surface, in real time, using an imitation substrate with integrated sensors. The direct incumbent.	Up to 13 sensing elements per holder; ion current density 0.0001 A/m² to 700 A/m² across three ranges [S40]
Hiden Analytical EQP series	Combined ion energy analyser and quadrupole mass spectrometer; resolves the ion energy distribution *per ion species*, which no optical inversion claims.	Mass ranges to 5000 atomic mass units; sampled through an orifice at one point [S41]

Neither publishes an accuracy figure, an uncertainty budget, or a price. Both vendor pages were opened and read [S40], [S41]. The Hiden page rejects some automated retrieval clients but serves normally to an ordinary browser request, and its published specifications (mass ranges, species resolution) are as described above. **Neither page carries an accuracy figure, an uncertainty budget, or a price.**

6.1.1 The pricing structure of this industry

With one exception, the plasma diagnostics industry does not publish prices. Impedans' Semion and Langmuir probe pages carry "Request Information" and no price in any currency [S40]; Tech-X's licensing page offers "a diverse range of software licensing options" with no licence types and no figures [S42]; COMSOL's licensing page describes six licence types and no prices [S43].

The exception is Plasma Controls, LLC, which publishes a full public price list [S44]. **Assessment:** That single exception is why this report states "with one exception" rather than "no vendor publishes prices". The distinction matters: an earlier research pass concluded no vendor published prices at all, and a later pass found the exception. The corrected statement is the one in this paragraph.

There is no commercial turnkey low-temperature-plasma Thomson scattering product. Searching for one returned no vendor offering; every system found in the literature is a bespoke laboratory build. **Assessment:** The absence of a price list for Thomson scattering is therefore a market-structure fact, not a research failure, *there is no product to price*.

6.2 Simulation codes that predict the quantity

Code	What it does	Stated limitation
hPIC / hPIC2	Fully kinetic electrostatic particle-in-cell for plasma–material interactions; produces ion energy–angle distributions at the wall. The closest-purposed forward model found.	Requires supercomputer-class resources: scaling tested to 262,144 cores [S45]. **Assessment:** No public source repository for the successor code hPIC2 was found: the developing laboratory's public organisation exposes only a metadata repository and a documentation repository *for hPIC2 specifically* [S46].
HPHall / Hall2De	Hybrid and axisymmetric Hall thruster discharge codes used for service-life and erosion prediction.	Anomalous cross-field transport is represented by a fitted mobility model rather than derived physics: [S2] records that HPHall "assumed Bohm's 1/B scaling for the cross-field mobility", while Hall2De replaced Bohm-driven transport with "a multi-variable spatial model of the anomalous collision frequency" that is empirically derived and piecewise in space.
EDIPIC-2D	Electrostatic particle-in-cell with secondary electron emission and ion velocity distribution outputs; the best-matched *open* sheath forward model.	Last repository push May 2024, i.e. approximately two years stale [S47].

Assessment: The critical observation is that every code in this table is a *forward* model. It predicts observables from an assumed plasma state. None inverts measurements to recover the state, and none returns a posterior. The proposed technology consumes this category rather than competing with it, which makes these codes candidate dependencies rather than rivals.

Cost is also decisive: a 2020 community review states that particle-in-cell simulations "are computationally expensive and therefore not yet adapted for thruster design optimization" [S2].

ADAM ASTROPLASH

7 Competitive Landscape

7.1 The category that actually competes

Bayesian fusion of plasma diagnostics is an established discipline in research operation in magnetic confinement fusion. It is not a gap waiting to be filled.

Framework	What it infers, from what	Does it target sheath ion energy flux?
Minerva at W7-X	Bayesian graphical models fusing **four** channels: interferometer, Thomson scattering, helium beam emission spectroscopy and X-ray imaging crystal spectrometers. Infers n_e, T_e, T_i profiles, plus Thomson calibration factors, as unknowns [S5].	**No**, verified by reading the paper. Its only mention of a material surface is empirical constraints at limiter/divertor positions introduced as *virtual observations*, i.e. a prior boundary constraint, not a sheath model [S5].
Integrated Data Analysis at ASDEX Upgrade	Bayesian combination of heterogeneous diagnostics for ill-posed inversion; applied since 2004 [S16]. **Assessment:** That date, and an earlier 1993 entry sometimes quoted alongside it, come from a personal background slide in a tutorial lecture rather than from an institutional record; the peer-reviewed carrier for the method is [S4].	**No**, verified. Scope is core and edge *profiles*; sheath and plasma–surface ion energy flux are not in scope [S17].
D-MIBAS at MAST-U	**The closest competitor found.** Multi-instrument Bayesian analysis fusing Langmuir probes, divertor Thomson scattering and spectrally filtered camera imaging, inferring T_e, n_e and neutral hydrogen density over a 2D poloidal cross-section [S6].	**Partial, and closer than it first appears.** It is multi-diagnostic Bayesian inference *in the plasma–surface region*, and it **does derive parallel heat and ion fluxes with uncertainty** in the flux tube [S6]. **Assessment:** This is drawn from the paper's derived-quantities discussion rather than its abstract, and could not be re-verified at source within this assessment because the full text exceeded automated retrieval limits; treat the flux derivation as established and its exact geometry as unconfirmed. What it does not do is resolve flux *arriving at the surface through a sheath*: it contains no sheath forward model, and it is not non-intrusive, since Langmuir probes are one of its three channels.

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Framework	What it infers, from what	Does it target sheath ion energy flux?
Hall thruster Bayesian calibration	Multifidelity Bayesian calibration of a coupled cathode, discharge and plume model; outputs the angular distribution of plume current density [S20].	**Partial.** Genuine physics-constrained Bayesian inference with calibrated uncertainty on a plasma device, but the surface-adjacent quantity is plume current density, not energy-resolved flux through a sheath.
Bayesian inference of ion velocity distribution from LIF	Bayesian model selection over analytic ion velocity distribution forms from laser-induced fluorescence spectra [S48].	**No**, single diagnostic, spatially local, no sheath. A component method the proposed framework would subsume, not a rival.

The differentiator holds, stated to the limit of its evidence. None of the five frameworks whose primary text was read for this report (Minerva/W7-X, IDA/ASDEX-U, D-MIBAS/MAST-U, the Hall thruster calibration work and the Bayesian LIF work) **infers ion energy flux at a surface via a sheath forward model**. **Assessment:** Those five were selected as the most-cited systems surfaced by the competitive research pass, not by a systematic literature survey with a recorded query set. The finding is therefore strong evidence about the leading frameworks and *not* a field-level negative. Establishing the stronger claim requires the structured prior-art search recommended in §11.2.

Assessment: D-MIBAS should be treated as the benchmark and the principal competitive threat. It already performs multi-diagnostic Bayesian inference in the plasma–surface region and publishes hard accuracy numbers (§7.2.2). Its two structural gaps are precisely the two things this project fixes, which is a strong story, but also means the same group is plausibly one publication away from closing them. **Assessment:** It already derives parallel heat and ion fluxes with uncertainty [S6], so the remaining distance is smaller than a first reading suggests.

7.2 Positioning matrix



Figure 7.1: Competitive positioning. Entities placeable on both axes appear in the plane; entities placeable on one axis appear as rug marks in the corresponding gutter; entities placeable on neither are named in the side panel with the reason. The subject technology appears on the channel axis only, because it has no accuracy result. Source: [S5], [S18], [S4], [S6], [S20], [S48], [S40], [S41], [S49], [S50], [S51], and [C] for the subject technology's channel count. Underlying data: Appendix F, Table F.2.

7.2.1 Axis 1: fused diagnostic channels

Placeable with cited counts: Minerva at JET approximately 10 diagnostic systems [S18]; Minerva at W7-X 4 [S5]; Integrated Data Analysis at ASDEX Upgrade 4 [S4]; the subject technology 4 **[C]**; D-MIBAS 3 instrument classes [S6]; Impedans Quantum 2 [S49]; and six single-channel commercial instruments [S40], [S41], [S50], [S51], matching Table F.2 **[A]**.

Assessment: On this axis the subject technology *ties* two established academic frameworks and is beaten by a third. Channel count alone does not differentiate it. The axis is useful only for showing that the commercial field clusters at one to two channels while the academic field spans three to ten, that is, that this project's real peer group is academic, not commercial.

7.2.2 Axis 2: published accuracy

Placeable: D-MIBAS at 6 %, 5 % and 30 % median absolute percentage error for T_e, n_e and neutral atomic hydrogen density respectively, against SOLPS-ITER synthetic ground truth [S6]; the Hall thruster calibration work within 10 % on held-out flow rates and pressures, with 9 % error extrapolating to on-orbit conditions and a greater than 50 % error reduction against prior work [S20].

The commercially strongest finding in this report, stated to the limit of its evidence. Every entity in this report with a published error bar is academic. No accuracy figure and no uncertainty budget for the ion energy quantity was found on any of the vendor pages listed in Appendix E, specifically [S40], [S41], [S52] and [S44]. **Assessment:** This is a finding about the pages checked, not a census of the industry: the process-control firms named in the side panel of Figure 7.1 were identified but their documentation was not individually audited, so "without exception" would overstate what was verified.

Assessment: Two cautions on how to use that finding. First, a group of academic frameworks (Minerva, Integrated Data Analysis and related systems) are absent from this axis because they report *per-shot posterior uncertainties*, which is methodologically stronger than a single error percentage, not weaker. Conflating "no single number" with "no rigour" would be wrong, and a domain reviewer will catch it. Second, the subject technology now carries a number on this axis, 3.99 % **[C]**, and it must be read with the weight its provenance allows: it is synthetic, self-reported and not peer reviewed, whereas every other point on the axis is published. It is plotted distinctly in Figure 7.1 for that reason.

7.2.3 The axis that was rejected

A price axis was attempted and abandoned. **Assessment:** With exactly one vendor publishing prices [S44] and every other quote-only, the number of competitors placeable on a price axis is one. An axis with one placeable point is not a positioning tool, and drawing one would have implied comparative pricing knowledge this report does not have.

7.3 Adjacent commercial activity

- **Applied Materials has acquired Impedans**, the maker of the incumbent intrusive ion energy analyser [S53]. This is corroborated independently by the recorded assignment of the corresponding patent from Impedans Ltd to Applied Materials Inc [S54]. The deal value was not disclosed [S53].
- **Lam Research sells plasma simulation into fabs** through its Semiverse portfolio, whose published product set includes high-fidelity modelling of plasma discharges alongside its virtual-fabrication tools [S55].

Assessment: The Impedans acquisition is the highest-consequence commercial fact in this report, and it cuts in two directions simultaneously. It removes the most obvious independent channel partner for a substrate-level ion energy product, and it validates that the largest semiconductor equipment vendor considers this measurement strategically worth owning. For a software framework seeking a hardware partner, the field of independent partners just narrowed.

A repository search for open-source competitors in this exact niche returned **zero repositories** for both `plasma+sheath+bayesian` and `OES+plasma+inversion` [S56]. **Assessment:** That establishes only that no repository was found by those two queries. Two multi-term repository searches returning zero is weak evidence, since such searches return nothing for most specific topics, and the report's own evidence base contains a counter-example, since the laboratory organisation inspected in Chapter 6 [S46] hosts repositories in adjacent territory that these queries do not surface. A documented multi-query sweep would be needed before any occupancy claim, and Chapter 11 explains why that should not be read as freedom to operate.

An open licensing decision is a commercial decision, and it is still open. The team reports the code not public and the licence an unresolved architecture decision record **[C]**. **Assessment:** Two consequences follow, and only the first is a drawback. **No result in this file is externally reproducible until that decision closes**, which is what a reviewer will ask about first. But the decision itself is a genuine fork in the business model, open-source-with-services against proprietary licence, and the report's own market evidence bears directly on it: at the 84 addressable organisations **[A]** counted in Chapter 5 the proprietary licence route is grant-scale, which is an argument for taking the fork seriously rather than defaulting. Deciding it is cheap, and it gates both reproducibility and the revenue model.

TEAM ASTROPHAGE

8 Evidence and Validation

8.1 What the team claims, separated from what is corroborated

8.1.1 Claims made by the team, not independently verified

All of the following carry [C] and none has been executed or reviewed by a third party:

- Validation is closed-loop against synthetic ground truth, with deliberate model mismatch between the data-generating and inverting models (different mesh, different sheath model, different collision set) specifically to avoid inverse crime [C].
- Results are to be reported at three tiers: T0 self-consistency, T1 inverse-crime/optimistic, and T2 honest/mismatched, with **T2 the only tier quoted publicly** [C].
- Instrument realism is specified to commercial part numbers, with enumerated noise including Poisson photon-counting statistics, microchannel plate gain distributions and per-channel calibration uncertainties of 6 % to 7 % [C].
- Asynchronous, phase-binned acquisition timing is modelled explicitly rather than assumed away [C].

8.1.2 Externally corroborated context for those claims

Assessment: The methodological commitments above map onto known failure modes in this literature rather than being idiosyncratic, which raises confidence that the team understands its own field even though nothing has been run.

Specifically, the closest comparable framework openly documents the very limitations this design anticipates. Minerva at W7-X reports that "comprehensive sampling from the joint posterior distribution is deferred to future investigations" because of model complexity, meaning it currently uses maximum a posteriori point solutions rather than full posterior sampling; that its smoothness constraints do not capture minor structures; and that its calibration factors differ appreciably depending on how much edge-region information is available [S5].

Assessment: That third limitation (sensitivity to edge/boundary information) is directly adjacent to the sheath region this project targets. It is the clearest technical evidence in the file that the proposed contribution addresses a real, acknowledged seam rather than an invented one.

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8.2 The validation boundary, stated explicitly

The benchmark has been executed and its numbers supplied. The team reports the closed-loop synthetic benchmark run to completion, with results at every declared tier [C]. **Assessment:** The validation boundary has moved, and it is worth being exact about where it now sits. It is no longer between *specified* and *run*. It is between *synthetic* and *real*: every figure below was produced against the project's own forward model, and nothing in this file has met laboratory data.

Tier	Mean rel. error	What it tests
T0, analytic	0.012 % [C]	self-consistency, closed form
T0, fluid model inverted	0.035 % [C]	self-consistency, numerical
T1, noise added	0.050 % [C]	optimistic/inverse-crime tier
T2, honest, mismatched	**3.99 % ± 2.76 %** [C]	the only tier quoted publicly

The T2 figure is a mean over 10 noise realisations, relative to the true Γ_E at the sealed evaluation point, itself an analytic estimate of 9.40 kW/m² [C]. By the project's own reporting rule the two supporting quantities

are quoted beside it: density −3.54 % and temperature 8.20 % [C]. **Assessment:** **The two orders of magnitude between T1 and T2 are the honest headline**, and the team publishing that ratio rather than the T1 number is the single most creditworthy disclosure in the file.

Calibration, which is the commercial claim. Of the four tiers above, only one number bears on whether anyone would buy this: whether the stated uncertainty is honest. The team reports **8 of 10 credible intervals covering at the 90 % level** for the weighted four-channel configuration, against **0 of 10** for naive fusion [C]. **Assessment:** The calibrated-uncertainty claim survives, and it survives on the metric that matters rather than on an accuracy percentage. Two qualifications belong with it, and the team states both: at $N = 10$ the binomial standard error is approximately 0.09 [C], so 8 of 10 and 9 of 10 are not distinguishable and nothing should rest on that difference; and 80 % against a 90 % target is marginal under-coverage, not over-coverage.

Assessment: The temperature term carries a second result the team reports against itself. The 8.20 % bias had been hypothesised as an electron energy distribution mismatch; it was traced instead to model discrepancy carried by the LIF channel, from a sampling node that is not the fluid model's sheath edge [C]. **That is a located, fixable modelling defect rather than a diffuse one**, which is the better of the two outcomes, and it was found because the benchmark was run honestly enough to expose it.

[DATA GAP 9] *claim needed:* The identifiability map: the eigenvalue spectrum, condition number, Cramér–Rao comparison and resolvable/degenerate partition of the operating space. *Why unavailable:* The team reports that the Fisher-information machinery exists but that none of these was computed, and that the n_0–T_e degeneracy ($\Gamma_i \propto n_0\sqrt{T_e}$) is argued analytically rather than measured [C]. The absence is stated by the team rather than concealed. *how to get it:* Compute the spectrum and publish the partition of the operating space into resolvable and degenerate regions. This is now the single highest-value open item in the file: R1, identifiability, is the joint highest-scored risk in the register and is the one supported by the least evidence of any risk in it

TEAM ASTROPH

9 Technology Readiness and Maturity

9.1 Official scale definitions

Four scales were consulted. Three were read directly and are quoted verbatim below; the ISO criteria are quoted from their permitted ECSS reproduction, not from ISO itself. **Assessment:** The four are also not fully independent (ISO 16290 states that it was produced taking due consideration of the NASA and DoD documents) so their agreement is corroboration, not four separate votes.

NASA NPR 7123.1D, Appendix E, effective 5 July 2023, carries a software-specific column [S57]:

- TRL 1 software: "Scientific knowledge generated underpinning basic properties of software architecture and mathematical formulation."
- TRL 2 software: "Practical application is identified but is speculative; no experimental proof or detailed analysis is available to support the conjecture. Basic properties of algorithms, representations, and concepts defined. Basic principles coded. **Experiments performed with synthetic data.**"
- TRL 3 software: "Development of limited functionality to validate critical properties and predictions using non-integrated software components."

US Department of Defense, Technology Readiness Assessment Guidebook, February 2025, Table 2-2 software [S58]: TRL 2 is "…limited to analytic studies using synthetic data"; TRL 3 extends to validating critical properties "using non-integrated software components and **partially representative data**".

European Commission, Horizon Europe General Annexes [S59]: "TRL 1 – Basic principles observed; TRL 2, Technology concept formulated; TRL 3 – Experimental proof of concept."

ISO 16290:2013 defines TRL 2 as requiring "No proof of concept yet" and TRL 3 as performance "demonstrated through analytical models supported by experimental data/characteristics". **Assessment:** These criteria are quoted from ECSS-E-AS-11C [S60], which reproduces ISO 16290:2013 Table 4-2 with ISO's written permission; the ISO document itself is paywalled and its catalogue record [S61] carries no level definitions. Note also that ISO 16290 states its own scope as applying "primarily to space system hardware", so it is the weakest of the four scales for a software project and is cited here only for convergence.

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9.2 Assignment

Assessment: Mapping the Context Block against the software-specific wording, which is the correct scale for a computational framework:

- **TRL 1 is fully met.** The architecture and the forward/inverse mathematical formulation are specified and documented at Baseline v1.0 **[C]**, verbatim NASA TRL 1 software [S57].
- **TRL 2 is met on all four of its clauses.** NASA's software TRL 2 has four clauses, and all four must be tested rather than the convenient ones [S57]. "Practical application is identified but is speculative", satisfied **[C]**. "Basic properties of algorithms, representations, and concepts defined", satisfied **[C]**. "Basic principles coded", **satisfied**: the team reports the framework coded and run **[C]**. "Experiments performed with synthetic data", **satisfied**: the team reports the closed-loop synthetic benchmark executed to completion **[C]**. **Assessment:** Note that this clause is met by the experiments having been *performed*. The official wording does not require them to have succeeded, so the level follows from execution irrespective of what the benchmark found.
- **TRL 3 is not met on any of the four scales.** Each requires experimental proof of concept, non-integrated component validation, or partially representative data. None exists.

Assigned readiness: TRL 2. TRL 1 is complete, and all four clauses of NASA's software TRL 2 are now reported met [C]. **Assessment:** The project has cleared the gate this assessment itself named: "TRL 2 completes when the closed-loop synthetic benchmark is executed" (§9.2). **Two qualifications belong with that assignment.** First, it rests on the team's report of execution and has not been independently verified by this assessment [C]; the run itself, not merely its results, is what an assessor should ask to see. Second, TRL 2 is a statement about work performed, not about work succeeding, so it carries no implication that the benchmark returned a useful figure. TRL 3 remains gated on validation against real laboratory data.

Two clean gates follow directly from the official wording:

1. **TRL 2 completes** when the closed-loop synthetic benchmark is executed, exactly NASA's "experiments performed with synthetic data" [S57]. **The team reports this gate passed** [C].
2. **TRL 3 is reached** when critical properties are validated against partially representative, i.e. real laboratory, data [S58]. **This gate remains open.**

9.3 Maturity dashboard

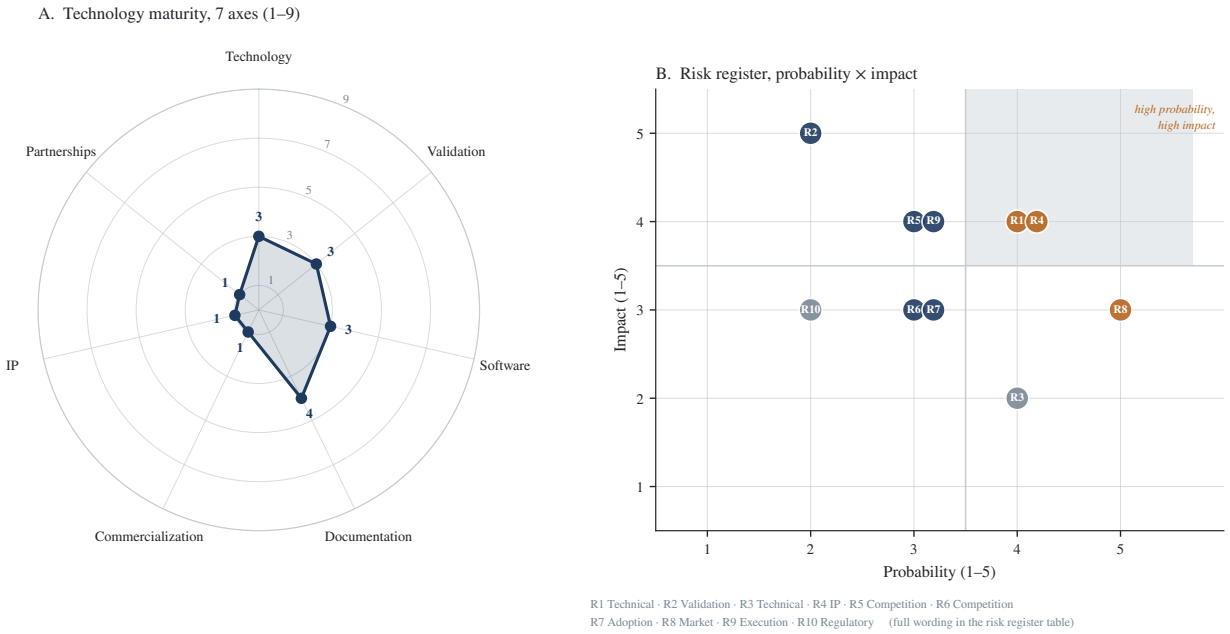


Figure 9.1: Panel A: technology maturity scored on seven axes, 1–9. Panel B: risk register plotted as probability against impact. Scores are the analyst's assessment, each carrying a written justification. No benchmark or comparison ring is drawn on Panel A, because no sourced definition exists of what a typical project scores on these axes. Source: analyst assessment throughout, with each score justified in Appendix F, Tables F.4 and F.5; no point on either panel is a measurement or a team claim.

Assessment: The dashboard's most informative feature is its asymmetry. Documentation scores 4 while every other axis scores 1 or 2 (Assessment), the project has specified far more than it has built, and built far more than it has validated. That shape is characteristic of a rigorous researcher working alone without laboratory access, and it is the correct shape to *start* from. It becomes a warning sign only if it persists after funding, because the axes that lag are precisely the ones money and partnership are supposed to move.

10 Commercialization Pathway and Business Model

10.1 What the comparables show

Each pathway below is tied to a named company with a cited figure.

Pathway	Comparable, with cited figure	Read-across
Federal R&D funding, then a modest exit	**Tech-X Corporation**, maker of the VSim plasma simulation code, took 157 federal assistance awards totalling $111,313,004 since fiscal 2008 plus 71 contracts totalling $23,427,957, retrieved by querying the awards API for that recipient [S38]. It operated from 1994 [S62] at "a staff of 40 people between employees and consultants" [S63]. It was then **acquired by Silvaco Group on 29 April 2025 for an aggregate purchase price of 8.0 million US dollars**, 3.9 million cash, 457,666 shares valued at 2.4 million, and contingent consideration with a maximum payout of 2.0 million and an estimated fair value of 1.7 million [S64].	**The single most directly comparable data point in this report.** An independent plasma-simulation software house, sustained on public research money for 31 years, realised 8.0 million at exit, of which only 3.01 million was identifiable technology and customer intangibles (developed technology 1.28 million, customer relationships 1.73 million) **[CALC]** from the purchase price allocation [S64].
Perpetual licence, no venture capital	**MathWorks**: "$1.5 billion in annual revenue", "profitable every year since founding" in 1984, privately held [S65].	Proves the licence model compounds without outside capital, but over decades, not quarters.
Employee ownership instead of exit	**Kitware**: became 100 percent employee-owned via an employee stock ownership plan approved 30 November 2020 [S66]; self-reports "more than 200 employees" and "more than $39 million consolidated revenue", describing its model as "innovation, not software, is our primary product" [S67].	A credible terminal structure for a research-software business that never wants an acquirer, and evidence that services-on-open-source scales further than licensing does at this size.
Public-market physics software	**Schrödinger**: fiscal 2025 software revenue 199.5 million US dollars, total revenue 255.9 million, net loss 103.3 million [S68].	The realistic public comparable for physics-simulation software, and note it is loss-making at that scale.

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Pathway	Comparable, with cited figure	Read-across
Strategic acquisition	**Oxford Instruments acquired Andor Technology** for 158.1 million pounds net cash consideration [S69]. **Nova acquired ancosys** for approximately 90 million US dollars in cash, a figure that *includes* a performance-based earnout of approximately 10 million [S70] and **Sentronics** for approximately 60 million [S71]. **Applied Materials acquired Impedans**, value undisclosed [S53].	**Two distinct exit patterns appear in this evidence, and they differ by an order of magnitude.** Instrumentation-with-software-attached produced £158.1 million (Andor, acquired as a supplier of cameras, microscope systems *and software* [S69]), approximately 90 million US dollars (ancosys) and approximately 60 million (Sentronics). **Standalone scientific software also has a demonstrated exit path** (Tech-X, in the row above) but it cleared 8.0 million [S64], roughly an order of magnitude lower **[CALC]**.
Ceiling case	**Synopsys acquired Ansys** at approximately 35 billion US dollars enterprise value, announced 16 January 2024, closed 17 July 2025 [S72], [S73].	**Assessment:** Included for completeness and immediately discounted: Ansys was a multi-decade, multi-physics platform business. It is not a comparable for a TRL 2 project and should not be presented as one.

10.2 Recommended pathway

Assessment: The evidence points to one sequencing, and it is not the one a software-first framing would suggest.

1. **Fund the next gate with non-dilutive research money, not equity.** The licence market is too small to justify dilution at this stage (§5.3), and the comparable funding instruments are well matched to the actual need. The three on-topic awards catalogued in §15.3 run from $300,000 to $399,999.56; the two closest analogues (funded projects to build a portable optical Thomson scattering system) were awarded $321,692.65 (award DE-AR0001165) [S36] and $399,999.56 (award DE-AR0001159) [S37].
2. **Pair with instrumentation if the goal is a larger exit.** Standalone scientific software in this domain *does* have a demonstrated exit path (Tech-X sold to Silvaco for 8.0 million US dollars [S64]) so the choice is not between an exit and no exit. It is between roughly 8 million and the £158.1-million-to-90-million-US-dollar range that instrument-plus-software businesses achieved [S69], [S70], [S53] **[CALC]**.
3. **Treat licence revenue as validation, not as the business.** At £3025 per seat per year [S32] across 84 organisations, licences signal adoption; they do not fund development.

Assessment: One structural constraint must be stated plainly, because it invalidates an otherwise obvious route and is frequently got wrong. United States SBIR, whose Phase I and Phase II ceilings of $323,090 and $2,153,927 [S74] would otherwise fit this project's needs precisely, is **not available to this applicant**. The regulation offers three alternative ownership routes, the first of which requires an awardee to be "more than 50% directly owned and controlled by one or more individuals (who are citizens or permanent resident aliens of the United States)"; the others permit ownership by other United States-owned small businesses or by multiple investment companies [S75]. **Assessment:** The ownership test is therefore not by itself decisive. What is decisive is place of performance: the administering agency states that "All funded work, including work done by consultants and contractors, needs to take place in the United States" [S76].

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Assessment: That second condition is the one that closes the obvious workaround. Incorporating a United States entity would not suffice, because the science could not then be performed in India. Using this route is a relocation and capitalisation-table decision, not a grant application. The realistic instruments for this applicant are domestic and are enumerated in Chapter 15.

The comparable that should anchor expectations. Tech-X ran the exact business model this project would adopt (plasma simulation software funded by public research money) for 31 years **[CALC]** (1994 to the April 2025 sale), drawing over 111 million US dollars in federal assistance [S38], and exited at 8.0 million [S64]. **Assessment:** A committee should size its expectations against that number rather than against the Ansys headline. The pathway is viable and durable; it is not venture-scale.

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11 Technology Moat and IP Strategy

11.1 What the patent search found

Eighteen patents were reviewed, with claim 1 read directly for fifteen of them and the remaining three recorded as not claim-verified **[A]**. **Assessment:** Five of the eighteen are reproduced in the table below, and a sixth [S21] is discussed in Chapter 4; the remaining twelve were reviewed and judged not material to the assessment, and are not individually cited here. The most relevant are:

Patent	Assignee, dates	Claim 1 scope, as read
US 12,362,159 B2	Lam Research; filed 15 March 2022, granted 15 July 2025 [S22]	**Claim 1, read in full:** receive a measurement of plasma sheath *voltage* and a value of the sheath's *total charge*, "wherein the first value is determined based on an optical signal sensed by an optical sensor"; determine sheath *capacitance* from those two; and modify an RF generator variable "until it is determined that the first capacitance is equal to" a predetermined capacitance. **The closest live art.** **Assessment:** An optical observable is already coupled to a physical sheath quantity inside a closed control loop.
US 11,908,668 B2	Impedans Ltd, assigned to Applied Materials 12 September 2025 [S54]	A wafer-like substrate carrying a four-grid retarding-field analyser with integrated battery and control circuitry. The incumbent intrusive instrument.
US 9,970,818 B2	Tokyo Electron; filed 31 October 2014, granted 15 May 2018 [S77]	Measure N line-integrated optical emission spectra along non-coincident rays and solve for the spatial emission distribution. **The nearest existing inverse-method claim**, but deterministic, single-diagnostic, no posterior.
US 5,347,460 A	IBM; granted 13 September 1994, **expired** [S78]	Optical emission spectroscopy for semiconductor process control. Its specification expressly uses a **Bayesian classifier** on optical emission data with priors updated by observed spectra.
EP 2122657 B1	Ruhr-Universität Bochum; granted 22 June 2011 [S79]	Controlling ion energy via multi-harmonic RF phase, that is, *actuation* of ion energy, not optical inference of it. **Assessment:** Cited only as evidence that university-origin IP exists in the adjacent sheath-control space. **No licensing is claimed**: the cited record lists an individual as the current assignee, and nothing on it evidences transfer to industry.

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11.2 Whitespace, phrased against the evidence

Assessment: Stated strictly, because absence of evidence is not evidence of absence: **no patent was retrieved, by the twenty-three searches listed below, claiming Bayesian inversion of multiple fused non-intrusive optical plasma diagnostics against a physics-based sheath forward model to produce a calibrated posterior over ion energy flux.** The closest retrieved items each occupy one axis only, deterministic inversion of a single channel [S77], a single optical observable driving a deterministic control loop [S22], or empirical multi-sensor regression with no physics model and no posterior.

The searches themselves. Twenty-three keyword searches were run over Google Patents and USPTO full text on 6 August 2026 **[A]**. **All twenty-three query strings are reproduced verbatim in Appendix C**, so the absence claim above can be reproduced or falsified by anyone.

The search had material limits, and they are disclosed here rather than buried: it was keyword-based and English-only; no Cooperative Patent Classification sweep was performed; Espacenet, PATENTSCOPE and Lens.org returned no usable results, so coverage is essentially Google Patents and USPTO full text; no Japanese, Korean or Chinese language searching was done, which is a material gap given the assignee profile; and applications published within roughly eighteen months are invisible by construction.

The finding that inverts the usual reading. Thin patent coverage here should be read as a **novelty risk**, not as an open field. The non-patent literature is dense with exactly this method, Bayesian integrated data analysis [S17], Bayesian inference of ion velocity distributions from LIF spectra [S48], multi-instrument Bayesian inference in a divertor [S6]. Dense non-patent prior art is a validity problem, and it is the most likely explanation for why the patent record looks empty.

Assessment: The practical consequence is that a literature-based prior-art search is a higher-value next step than further patent keyword searching. If a claim is to be pursued, the defensible combination on this evidence is not the Bayesian method or the channel set alone, but those *plus a control action conditioned on a posterior credible interval*, because that is what [S22] shows the industry actually claims, and it is the element the academic literature does not contain.

11.3 Filing costs

United States Patent and Trademark Office fees, effective 19 January 2025: a provisional application filing fee is $325 undiscounted, $130 for a small entity and $65 for a micro entity; a nonprovisional filing, search and examination together total $2000, $800 and $400 respectively [S80].

For an international filing at the United States receiving office electing the United States searching authority, the international filing fee is $1667, transmittal $285 and search $2400 [S81], giving approximately $4352 in government fees before attorney costs **[CALC]**. **Assessment:** That total uses *undiscounted* amounts throughout, unlike the USPTO figures above, which are given at all three entity tiers; small- and micro-entity reductions apply to some PCT components and would lower it.

[DATA GAP 10] *claim needed:* Whether the applicant qualifies for the WIPO 90 percent fee reduction available to applicants from certain states. *Why unavailable:* The reduction exists and is documented, but eligibility depends on a current list of qualifying states that was not checked in this research pass. *how to get it:* Check the applicant's nationality and residence against the current WIPO schedule of fee reductions, which is published and free to consult

12 Risk Analysis

Each risk below is tied to a specific finding earlier in this report rather than to a generic category. Probability and impact are scored 1–5 and plotted in Figure 9.1, Panel B.

ID	Category	Risk and its evidential basis	P	I	Score
R1	Technical	Identifiability limits mean ion energy flux cannot be resolved across significant regions of operating space. The team's own design reports identifiability via Fisher information and null-space analysis and commits to stating what it cannot resolve [C], so partial non-identifiability is anticipated by the team itself. Unchanged, and now the least-evidenced high risk in this register: the team reports that no eigenvalue spectrum, condition number, Cramér–Rao comparison or resolvable/degenerate partition was computed, and that the n_0–T_e degeneracy is argued analytically rather than measured [C].	4	4	16 [CALC]
R2	Validation	T2 accuracy lands too wide to be decision-useful. Probability lowered from 3 to 2 on a measured result: T2 is reported at 3.99 % [C], inside the 6 %, 5 % and 30 % triple the nearest published benchmark reports for T_e, n_e and neutral atomic hydrogen density [S6]. Not lowered further, because the result is synthetic, self-reported, and covers 8 of 10 rather than 9 of 10 [C].	2	5	10 [CALC]
R3	Technical	The projected compute budget is wrong because it rests on an unmeasured GPU throughput [C]. Score unchanged, but the risk is now narrower: G-1.4 is reported met for the L1 fluid solver only, at 86 ms per solve against an estimate of about 1 s, while the L2 particle-push throughput was not measured and is carried forward to G-2 [C]. High likelihood, low consequence *because* the team gates it and reports which half it has closed.	4	2	8 [CALC]
R4	IP	Novelty defeated by non-patent prior art rather than by patents. Bayesian multi-diagnostic fusion was published in the peer-reviewed fusion literature by 2010 [S4] and has been published in electric propulsion [S20]. See §11.2.	4	4	16 [CALC]

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ID	Category	Risk and its evidential basis	P	I	Score
R5	Competition	D-MIBAS closes both of its gaps first. It already performs multi-diagnostic Bayesian inference in the plasma–surface region with published accuracy [S6]; its two structural gaps are the two this project fixes.	3	4	12 **[CALC]**
R6	Competition	An incumbent absorbs the capability. Applied Materials has acquired the maker of the incumbent instrument [S53]; Lam already sells plasma simulation into fabs [S55].	3	3	9 **[CALC]**
R7	Adoption	Buyers have no procurement precedent for paying for calibrated uncertainty. No accuracy or uncertainty figure was found on any of the four vendor pages checked (§7.2.2). **Assessment:** Scored 3 rather than 4 because that is an observation about four pages, not a census of procurement practice.	3	3	9 **[CALC]**
R8	Market	Licence revenue alone cannot sustain a company: 84 organisations **[CALC]** at the one published comparable rate yield £254,100 to £762,300 per year **[CALC]** (§5.3), grant-scale income, before any discount for non-adoption.	5	3	15 **[CALC]**
R9	Execution	No access to a validation facility. No plasma diagnostics group is named on any of the three Indian institutional pages checked [S82], [S83], [S84]. **Assessment:** Scored at probability 3 rather than 4 because a concrete in-country route does exist and is identified in Chapter 14: IPR Gandhinagar has a named plasma diagnostics group [S85] and hosts AIC-Plasmatech, a Department of Atomic Energy-owned deep-tech incubator [S86]. The risk is that the route is not secured, not that none exists.	3	4	12 **[CALC]**
R10	Regulatory	Export control or dual-use restriction on fusion and electric propulsion diagnostics. **Assessment:** Plausible given the application domains, but **no specific control classification was verified**; the evidence for this risk is materially weaker than for the other nine and it is scored low in consequence.	2	3	6 **[CALC]**

[DATA GAP 11] *claim needed:* Whether plasma diagnostic inversion software for fusion and electric propulsion applications falls under any export control or dual-use classification. *Why unavailable:* Assessed as plausible from the application domains, but no specific classification was checked against any control list in this research pass. *how to get it:* A review against the relevant national export control list, or an opinion from a trade compliance adviser, before any international collaboration or code release

Assessment: Two risks score 16 **[CALC]**: R1 identifiability and R4 novelty. **R1 is a genuine capability risk**, if the flux cannot be resolved across much of the operating space, the method does not deliver its output. R4 is different in kind: it is the risk that the method works and cannot be defended. R9, facility

access, scores 12 **[CALC]** and is the most tractable of the three, because a specific route is already identified (Chapter 14). That is a materially different risk profile from a typical deep-tech project, and it argues for spending early money on partnership and prior-art clearance rather than on engineering.

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13 National and Strategic Importance

Assessment: This chapter is the easiest in any dossier to fill with unsupported national-interest language. Every claim below is therefore tied to a named programme and an official document, and programmes whose budgets are not published are recorded as such rather than estimated.

13.1 India

- **India Semiconductor Mission, Semicon 2.0.** Approved with a fiscal outlay of 127,500 crore rupees, structured in six pillars, of which the second is **Machines and Materials**, explicitly targeting "companies involved in the manufacturing and R&D of the machines… essential for manufacturing semiconductors" [S87]. Capital expenditure incentive rates are 40 % for CMOS silicon fabs, 35 % for other fabs, 35 % for advanced packaging and 25 % for conventional packaging [S87]. The predecessor programme carried an outlay of 76,000 crore rupees [S88].

  Assessment: This is the strongest strategic hook available. Domestic semiconductor *equipment and instrumentation* is now a named, funded national priority rather than an implied one, and process-control instrumentation is squarely within the Machines and Materials pillar's stated scope.

- **ISRO electric propulsion.** ISRO completed a 1,000-hour life test of a 300 mN stationary plasma thruster on 27 March 2025, at full power 5.4 kW on xenon, with the release stating that "the erosion of the electrode liners was periodically monitored" [S89]. The system is "proposed to be inducted and validated in the upcoming Technology Demonstration Satellite (TDS-01)" [S89].

  Assessment: Note what was being monitored, electrode liner erosion. That is precisely the quantity this technology infers, in a domestic programme, at a named facility, with a completed test campaign, the most direct national application in the file.

- **ITER-India.** India delivers nine packages to ITER as one of seven partners, contributing 9 % of in-kind contributions, and package nine is **"Selected Diagnostics, advanced diagnostic tools to monitor and analyze plasma behavior"** [S90].

- **National Quantum Mission** is the closest comparable deep-tech funding structure: 6003.65 crore rupees over eight years, with four thematic hubs of which one is **Quantum Sensing and Metrology** at IIT Bombay [S91].

  Assessment: Cited as structural precedent rather than as a funding route: it demonstrates that Indian national missions do fund *measurement science* as a first-class deliverable with a startup channel attached.

[DATA GAP 12] *claim needed:* Any Indian national mission dedicated to fusion or plasma technology, with a published outlay. *Why unavailable:* No such standalone mission was found on any Department of Atomic Energy, Department of Science and Technology, or Press Information Bureau page that could be opened; India's fusion effort is delivered institutionally through the Institute for Plasma Research rather than as a funded mission. *how to get it:* Direct enquiry to the Department of Atomic Energy, or review of the DAE annual report, which may state programme-level allocations

13.2 International

- **ITER** operates a diagnostic set of **60 instruments measuring 101 parameters** across seven categories, of which one is optical (explicitly including Thomson scattering and interferometers) and another is plasma-facing, covering chamber conditions, temperature and *erosion* [S92].

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- **United States Department of Energy** Fusion Energy Sciences names “Fusion Materials and Internal Components” as one of four research areas [S93]. Its Milestone-Based Fusion Development Program obligated 46 million US dollars for an initial 18 months against a total authorisation of 415 million through fiscal 2027, across eight awardees [S94].
- **NIST CHIPS Metrology Program** defines seven Grand Challenges, of which the first is metrology for materials and processing and the fifth is digital twins and semiconductor manufacturing process models [S95].

  Assessment: The intersection of those two Grand Challenges is the cleanest international articulation of this technology’s value proposition available from any official source.
- **EUROfusion** received 549.4 million euros from Euratom for 2021–2025, with approximately 450 million in member-state matching, across 31 research organisations [S96].

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14 Partnership Landscape

Assessment: Partnership claims are the easiest place in a dossier to imply relationships that do not exist. Nothing below is a relationship. All are verified *targets*, and the distinction is maintained explicitly.

14.1 India

- **Institute for Plasma Research, Gandhinagar**, the only Indian organisation with a verified, named plasma diagnostics group. Its SST-1 division lists a Plasma Diagnostics group with spectroscopy, laser diagnostics, imaging and X-ray, and microwave and infrared thermography sub-units [S85]. Its Advanced Plasma Studies division separately lists **Low Power Thruster and High Power Thruster** activity [S97].

  Assessment: This single institution covers both the diagnostic capability and the electric propulsion application. It is the highest-value partnership target in the file by a wide margin.

- **AIC-Plasmatech**, an Atal Incubation Centre established December 2023 as a Section 8 company **wholly owned by the Department of Atomic Energy**, sited at the Institute for Plasma Research, existing to "foster the growth of deeptech startups", offering mentoring by IPR scientists and access to IPR technologies and infrastructure [S86].

  Assessment: A deep-tech incubator, owned by the atomic energy department, located inside the one institution with the required laboratory; this is the single most actionable item in the entire dossier, and it directly addresses risk R9.

Explicitly not claimed. Institutional pages for the aerospace engineering departments at IIT Madras and IISc, and the physics department at IIT Bombay, were checked and **none names a plasma, electric propulsion or plasma diagnostics laboratory** [S82], [S83], [S84]. Individual faculty may work in the area, but no institutional page confirms it, so no partnership claim is made.

14.2 International

- **Georgia Tech High-Power Electric Propulsion Laboratory** explicitly includes "non-invasive plasma diagnostics" and the laboratory operates laser Thomson scattering systems [S34]. **Assessment:** The strongest single international validation-partner match found.

- **DIFFER**, Netherlands, operates the Magnum-PSI linear plasma device for plasma–material interaction under fusion-relevant conditions, with groups in Plasma Edge Physics and Diagnostics and Plasma Material Interactions [S98]. **Assessment:** The best-matched plasma–surface-boundary facility internationally.

- **Max Planck Institute for Plasma Physics**, operates ASDEX Upgrade and Wendelstein 7-X, with divisions in plasma diagnostics and plasma–wall interaction [S99]. **Assessment:** Also the home of the closest prior art, which makes it simultaneously the best-informed potential collaborator and the most capable potential competitor.

[DATA GAP 13] *claim needed:* Verified capability pages for PPPL, the University of Michigan Plasmadynamics and Electric Propulsion Laboratory, LPP at École Polytechnique, and the Air Force Research Laboratory. *Why unavailable:* All four sites rejected automated access with HTTP 403 or a Cloudflare challenge; these are real and well-known groups but no official page could be opened to verify their current programmes. *how to get it:* Manual browser retrieval of the four sites, or direct contact with the named laboratories

15 Investment Readiness

15.1 Current position

Stated from the Context Block, without inference: no company formed, no patent filed, no hardware built, no real experimental data acquired, no benchmark executed to completion, and documents at Baseline v1.0 dated August 2026 [C].

[DATA GAP 14] *claim needed:* Any current funding, budget, runway, team size or institutional support figure for the project. *Why unavailable:* The Context Block states no funding information, and no company or financial record exists to search. *how to get it:* Direct disclosure from the applicant, which is a reasonable request at the first committee meeting

15.2 Instruments actually available to this applicant

Assessment: The applicant appears to be India-based, which materially changes the funding routes available and rules out the most-cited one.

Instrument	Published amount	Fit at TRL 1–2
Startup India Seed Fund Scheme	Up to 20 lakh rupees as grant for proof of concept and prototype development, plus up to 50 lakh rupees for market entry via convertible debentures or debt [S100]	**Best fit.** Requires DPIIT recognition and incorporation within two years. Preference sectors include space and defence [S100].
NIDHI-PRAYAS	Up to 10 lakh rupees to an innovator or startup [S101]	Sector-agnostic, routed through a partner incubator, for which AIC-Plasmatech is the natural host [S86].
Technology Development Board	Up to 70 % of project cost, capped at 3.5 crore rupees for projects under 5 crore [S102]	**Not available yet.** Eligibility requires TRL above 6 [S102]; this project is at TRL 2 (§9.2).
United States SBIR	Phase I up to $323,090; Phase II up to $2,153,927 [S74]	**Not available.** The decisive bar is place of performance: all funded work must be performed in the United States [S76]. Ownership tests also apply but admit three alternative routes [S75]. Listed to document why it is excluded, since it is the most-cited instrument in deep-tech dossiers and the exclusion is frequently missed.

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Instrument	Published amount	Fit at TRL 1–2
BIRAC	n/a	**Not applicable [A].** BIRAC is a public sector enterprise of the Department of Biotechnology and its published remit is biotechnology; a plasma diagnostics project falls outside it. Recorded because it is frequently listed in error. **Assessment:** This is a reading of the body's stated mandate, not a formal eligibility determination.

15.3 What the next stage costs, benchmarked against funded projects

Assessment: Rather than estimate a budget, this section anchors on what comparable work has actually been awarded.

- Two funded projects to build a portable or transportable optical Thomson scattering system were awarded $321,692.65 (award DE-AR0001165, UC San Diego) [S36] and $399,999.56 (award DE-AR0001159, Caltech) [S37].
- A National Science Foundation award for a continuous-wave laser Thomson scattering diagnostic for low-temperature plasmas was $300,000 (award 2010466) [S35].
- One arm of a collaborative award specifically for understanding *sheath formation* was $318,198 (award 2108636, University of San Diego) [S103]. **Assessment:** This is one institution's obligation, not the project total.
- Component benchmarks from awarded federal contracts: a tunable diode laser with digital controller at $23,882 to 46,040; Andor intensified cameras at $29,310 to 48,860; a complete Nd:YAG laser with third harmonic generator, seeder, installation and training at $129,200 **[A]**. Each is an individual awarded federal contract **[A]**, identified by contract number: the laser range spans N0018917PZ177 through 1333ND24PNB680516, the camera range NNG10HJ88P through DOCSB134110SU0450, and the Nd:YAG award is NNG14HD00P [S38]. **Assessment:** These are government purchase amounts including shipping and, in some cases, accessories; they are not list prices, and no vendor in this category publishes one.
- Cloud compute comparison: published on-demand rates of $1.09 per GPU-hour for an A6000 and $0.69 for a Quadro RTX 6000 [S104]. **Assessment:** These are the closest published analogues to the workstation-class card the project specifies; NVIDIA publishes no manufacturer suggested retail price for that card [S105].
- European comparables for the same class of work: an EU fellowship to recover an unmeasurable physical quantity by iteratively matching a forward simulation against camera data (methodologically the closest funded analogue found) was awarded 171,399.36 euros [S106], and a laboratory programme using laser-induced fluorescence and Thomson scattering was awarded 226,420.56 euros [S107].

15.3.1 Facility cost, and an argument for sequencing

Published rates from a university nanofabrication facility show the same tool charged at very different rates by user class: an inductively coupled plasma etcher at $55 per hour for an academic user against $352 per hour for an industrial user, and a reactive ion etcher at $37 against $233 [S108]. Industrial rates therefore run approximately 6.4 and 6.3 times academic rates on those two tools **[CALC]**.

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Assessment: **This has a direct sequencing consequence.** *Tool-hour rates* at this one facility run roughly six times higher for an industrial user than an academic one **[CALC]**. That is a rate comparison on two etch tools at a single site, not a campaign cost (labour, materials and overhead are not included) but the direction is unambiguous. For a project whose licence market is small (§5.3), that argues strongly for **completing laboratory validation before incorporating**, not after. Incorporating early in order to look investable would materially increase the cost of becoming investable.

Assessment: The defensible envelope for a funded plasma diagnostics validation programme, bracketed by the three awards above that are both on-topic and whole-project (the NSF continuous-wave Thomson scattering award and the two ARPA-E portable Thomson scattering awards) is **$300,000 to $399,999.56**. The immediate next gate (the T0/T1/T2 benchmark) requires none of that, because it needs no hardware. It needs compute and time only. **That asymmetry is the investment recommendation: the cheapest possible action is also the one that resolves the most uncertainty.**

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16 Bibliometric Market Justification

16.1 Method

Counts were retrieved from OpenAlex, one database only, restricted to journal articles and conference papers, on 6 August 2026 [S109]. The type restriction is deliberate: unrestricted counts show a spurious 2026 spike caused by a shift in record-type mix toward preprints, datasets and software rather than by publication growth. 2026 is a partial year and is excluded from every fit.

Assessment: Two limits should be stated. All series come from a single database, so there is no cross-database validation; and an attempt to obtain a second series from Semantic Scholar failed on rate limiting, so no second source exists to check these against.

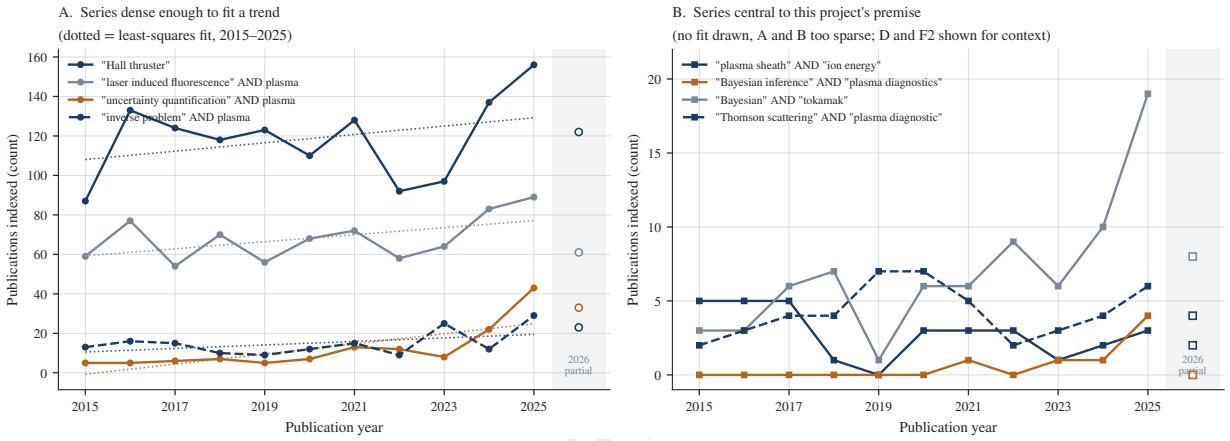


Figure 16.1: Publication counts by year. Panel A shows the four series dense enough to support a least-squares fit; Panel B shows the series central to this project's premise. No fit is drawn on Panel B because series A and B are too sparse to support one; series F2 is denser and is shown alongside them for context rather than as an unfittable series. The shaded band marks 2026 as a partial year, excluded from all fits. Source: [S109]. Underlying data: Appendix F, Table F.1.

16.2 What the counts show

Least-squares fits over the complete years 2015–2025 **[CALC]**:

- "Hall thruster": 87 in 2015 to 156 in 2025, slope +2.12 publications per year **[CALC]**.
- "uncertainty quantification" and plasma: 5 to 43, slope +2.56 per year, the steepest growth of any series measured **[CALC]**.
- "laser induced fluorescence" and plasma: 59 to 89, slope +1.78 per year **[CALC]**.
- "inverse problem" and plasma: 13 to 29, slope +0.89 per year **[CALC]**. **Assessment:** This series is visibly lumpy and should be described as noisy rather than as a clean trend.

16.3 The finding that cuts against the premise

The two series closest to this project's premise are the two thinnest in the literature. "plasma sheath" with "ion energy" returned 5 records in 2015 and 3 in 2025, flat to declining, with a year at zero [S109]. "Bayesian inference" with "plasma diagnostics" returned 7 records across 2015–2025 combined, summing column B of Table F.1 **[CALC]**.

Assessment: There are two honest readings and the committee should hold both. Read favourably, this is whitespace: very few groups work on exactly this intersection, which is consistent with the competitive finding that no framework occupies the position (Chapter 7). Read unfavourably, a field that is not growing may be one the community has found uninteresting or intractable. The evidence in this report does not settle between those readings, and any dossier claiming the intersection is a "growing field" is not supported by these counts.

Assessment: The defensible framing is the adjacent one: the *methodological* series is growing fastest of all measured (uncertainty quantification in plasma at +2.56 publications per year **[CALC]**) while the specific intersection remains unoccupied. That is a real and citable tailwind for the method, without over-claiming about the niche.

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17 Future Vision

Projection (scenario): Everything in this chapter is scenario, conditional on gates that have not been passed. No claim here should be read as a plan of record.

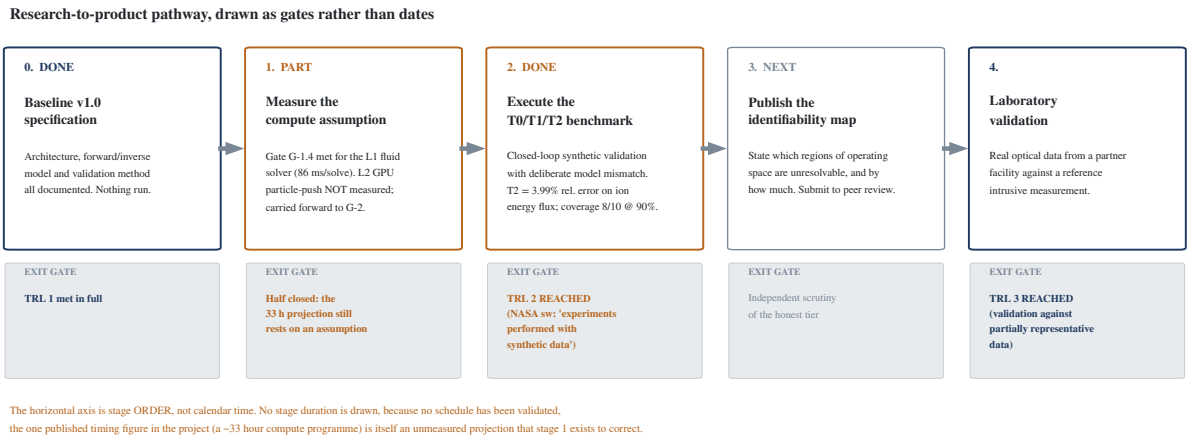


Figure 17.1: The pathway drawn as gates rather than dates. Stage durations are deliberately not drawn, because no schedule has been validated. Source: **[C]** for stage content; [S57] and [S58] for the TRL gate definitions.

Projection (scenario): If the T0/T1/T2 benchmark executes and the T2 tier lands within the range published by the nearest comparable framework (6 %, 5 % and 30 % median absolute error on T_e, n_e and neutral atomic hydrogen density respectively [S6], a range whose upper member is far wider than its lower ones) then the project would hold the first calibrated uncertainty figure for a non-intrusive sheath ion energy flux estimate *among the frameworks read for this report*, and the assumption embedded in that sentence is that mismatched-model inversion of a sheath quantity behaves comparably to matched-model inversion of volumetric quantities, which is not established.

Projection (scenario): If laboratory validation follows at a partner facility, the project would reach TRL 3 on the three software-bearing scales [S57], [S58], [S59] (ISO 16290 being scoped primarily to space hardware and cited only for convergence (§9.1)) assuming access to a facility with simultaneous optical access on all four channels, which has not been secured.

Assessment: The realistic terminal outcomes, ranked by the evidence in this report, are two, and both are modest.

1. **Acquisition by an instrumentation vendor** as a software layer on a diagnostic product. This is the only exit pattern observed anywhere in this domain [S69], [S70], [S53]. The directly comparable transaction (an independent plasma-simulation software house after three decades of federal funding) cleared 8.0 million US dollars [S64].

2. **A long-lived research-funded independent**, which is what Tech-X was for 31 years before that sale [S62], [S38], or what Kitware remains under employee ownership at over 39 million US dollars of consolidated revenue [S67].

Assessment: A standalone venture-scale software outcome is not supported by anything in this report: not by the market arithmetic in §5.3, not by the exit comparables, and not by the absence of an analyst category (§5.1). A committee should fund this as research infrastructure with a plausible modest exit, not as a venture bet.

A Evidence Matrix

The principal numeric claims in this report, with tag and source. **Assessment:** This matrix is comprehensive for the body chapters' load-bearing figures but is **not** an exhaustive index of every numeral in the document; ratios, dates, patent numbers and figure annotations are tagged in place rather than repeated here.

Ch.	Claim	Tag	Source
1	Langmuir probe expected to underestimate density by ~50% at stated reference conditions	[S]	[S1]
1	Density at probe surface one fifth of unperturbed; T_e 40% low	[S]	[S7] (itself citing an earlier work)
1	80% depletion	[CALC]	$1 - 1/5$, from the above
1	EEPF error under 3% requires $R_c/R_{p,\min} < 0.01$	[S]	[S8]
1	Three Recommended Practice papers, one 2017 issue	[S]	[S10]
1	NEXT: 50,170 h, 902 kg xenon, 34.9 MN·s	[S]	[S3]
1	8 years 3 months elapsed	[CALC]	June 2005 to September 2013
1	Qualification requirement 450 kg; ~22,000 h; lifetimes 10,000–100,000 h	[S]	[S12]
1	HERMeS design life exceeding 50 kh; erosion varies 76–300% and 40%; keeper erosion reduced 84%	[S]	[S13]
1	HERMeS LDWT 3,570 h; prior tests 1,700 h and 7×200 h	[S]	[S14]
1	ITER divertor 10 and 20 MW/m^2; 54 cassettes, ten tonnes	[S]	[S15]
3	Thomson 0.16 pe/shot at $n_e = 10^{17}$ m^{-3}	[C]	Team claim
3	Interferometry floor 8.4×10^{15} m^{-3} over 400 mm	[C]	Team claim
3	Calibration uncertainties 6–7% per channel	[C]	Team claim
3	~33 h on one RTX A4000	[C]	Team claim; a projection, not a runtime, gated for day-one measurement (Gate G-1.4)
5	Wafer fab equipment $115.7 bn (2025), $135.2 bn (2027); total equipment $133 bn	[S]	[S24]
5	Fusion funding $4.48 bn in 12 months; $14.24 bn to date = $13.26 bn private + $0.98 bn public	[S]	[S25]
5	Space propulsion $10.21 bn (2024) to $20.02 bn (2030), 11.9% CAGR	[S]	[S26]
5	Satellite propulsion $2.60 bn (2024) to $5.19 bn (2030)	[S]	[S27]
5	Rejected figure: EP-satellite market $13.03 bn (2025)	[S]	[S28]
5	Rejected figure: identical $13.03 bn for metrology market	[S]	[S29]
5	Rejected series: $1.86 bn (2025), $1.28 bn (2026), $3.09 bn (2034) at 6.56%	[S]	[S30]
5	31% decline and ~11.7% implied rate	[CALC]	from the figures on that page
5	Report-mill spread $67.8 m against $485 m	[CALC]	ratio of two mill figures; mills not cited
5	46 fusion companies, 21 research partners (archived snapshot 2026-05-16)	[S]	[S33]

TEAM ASTROPHAGE

Ch.	Claim	Tag	Source
5	10 etch OEMs (named); 7 EP facilities named, of which **1** confirmed from a primary facility page	[A]	enumeration; [S34] is the confirmed facility
5	Serviceable market = 84	[CALC]	46 + 21 + 10 + 7, from the dated snapshot [S33]
5	COMSOL seat £3,025/year	[S]	[S32]
5	Licence value £254,100 / £508,200 / £762,300	[CALC]	$84 \times 3025 \times \{1, 2, 3\}$
5	Diagnostic contracts $16,240–$86,080	[S]	[S38]
5	RPA $2,950 probe only, $4,450 with cables	[S]	[S39]
6	Semion ion current density 0.0001–700 A/m^2; 13 sensing elements	[S]	[S40]
6	Hiden EQP mass range to 5,000 amu	[S]	[S41]
6	hPIC scaling to 262,144 cores	[S]	[S45]
6	EDIPIC-2D last push May 2024	[S]	[S47]
6	hPIC2 source not found in the public organisation	[S]	[S46]
7	Minerva W7-X fuses 4 channels	[S]	[S5]
7	Minerva JET ~10 diagnostic systems	[S]	[S18]
7	IDA ASDEX Upgrade 4 channels	[S]	[S4]
7	D-MIBAS 6%, 5%, 30% median absolute error	[S]	[S6]
7	Hall thruster model within 10%, 9% extrapolation, >50% reduction	[S]	[S20]
7	Six single-channel commercial instruments	[A]	count of Table F.2
7	Zero repositories returned by two queries	[S]	[S56]
9	NASA software TRL 1/2/3 wording	[S]	[S57]
9	TRL 2: all four software clauses reported met	[C]	Team report of benchmark execution; not independently verified
9	DoD software TRL 2/3 wording	[S]	[S58]
9	EC TRL 1/2/3 wording	[S]	[S59]
9	ISO TRL 2/3 criteria (via the permitted ECSS reproduction)	[S]	[S60], [S61]
9	Maturity scores 1–9 on seven axes	Assessment	analyst judgement, each justified in Table F.4
10	Tech-X $111,313,004 assistance, $23,427,957 contracts	[S]	[S38]
10	Tech-X “a staff of 40 people”	[S]	[S63]
10	Silvaco acquired Tech-X for $8.0 m aggregate: $3.9 m cash, 457,666 shares at $2.4 m, contingent max $2.0 m (fair value $1.7 m)	[S]	[S64]
10	Identifiable intangibles $3.01 m (technology $1.28 m + customer relationships $1.73 m)	[CALC]	purchase price allocation in [S64]
10	Kitware 200+ employees, $39 m consolidated revenue	[S]	[S67]
10	Kitware 100% employee-owned, approved 30 Nov 2020	[S]	[S66]
10	MathWorks $1.5 bn revenue	[S]	[S65]
10	Schrödinger software revenue $199.5 m, total $255.9 m, net loss $103.3 m	[S]	[S68]
10	Andor £158.1 m net cash consideration	[S]	[S69]
10	ancosys ~$90 m + $10 m earnout	[S]	[S70]
10	Sentronics ~$60 m	[S]	[S71]
10	Impedans acquired by Applied Materials, value undisclosed	[S]	[S53]
10	Synopsys/Ansys ~$35 bn announced, closed 17 July 2025	[S]	[S72], [S73]

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Ch.	Claim	Tag	Source
11	Eighteen patents reviewed, claim 1 read for fifteen, 23 searches run	[A]	counts of the patent review
11	USPTO provisional $325 / $130 / $65; nonprovisional $2,000 / $800 / $400	[S]	[S80]
11	PCT $1,667 + $285 + $2,400	[S]	[S81]
11	PCT ~$4,352 government fees	[CALC]	1667 + 285 + 2400
12	Risk scores 16, 15, 12, 9, 8, 6	[CALC]	probability × impact
13	Semicon 2.0 INR 127,500 crore; capex rates 40% / 35% / 35% / 25%	[S]	[S87]
13	Semicon 1.0 INR 76,000 crore	[S]	[S88]
13	ISRO 1,000 h test, 300 mN, 5.4 kW, 27 March 2025	[S]	[S89]
13	ITER-India 9 packages, 9% of in-kind	[S]	[S90]
13	NQM INR 6,003.65 crore, four hubs	[S]	[S91]
13	ITER 60 instruments, 101 parameters	[S]	[S92]
13	DOE $46 m obligated, $415 m authorised, 8 awardees	[S]	[S94]
13	EUROfusion EUR 549.4 m plus ~EUR 450 m, 31 organisations	[S]	[S96]
13	NIST seven Grand Challenges	[S]	[S95]
15	SISFS INR 20 lakh / INR 50 lakh	[S]	[S100]
15	NIDHI-PRAYAS INR 10 lakh	[S]	[S101]
15	TDB 70%, INR 3.5 crore, TRL above 6	[S]	[S102]
15	SBIR $323,090 / $2,153,927	[S]	[S74]
15	SBIR place-of-performance and ownership rules	[S]	[S76], [S75]
15	Thomson scattering projects $321,692.65 and $399,999.56	[S]	[S36], [S37]
15	NSF $300,000 (award 2010466)	[S]	[S35]
15	Sheath formation award $318,198 (award 2108636)	[S]	[S103]
15	Laser $23,882–$46,040; ICCD $29,310–$48,860; Nd:YAG $129,200	[S]	[S38]
15	CORDIS awards EUR 171,399.36 and EUR 226,420.56	[S]	[S106], [S107]
15	Cornell facility $55 vs $352/hr and $37 vs $233/hr	[S]	[S108]
15	Industrial-to-academic ratios 6.4 and 6.3	[CALC]	352/55 and 233/37
15	GPU $1.09/hr A6000, $0.69/hr RTX 6000	[S]	[S104]
15	No published MSRP for the RTX A4000	[S]	[S105]
15	Validation envelope $300,000–$399,999.56	[CALC]	smallest and largest of the three on-topic whole-project awards
16	Publication counts by year and series	[S]	[S109]
16	Slopes +2.12, +2.56, +1.78, +0.89 per year	[CALC]	least-squares fit on Table F.1, 2015–2025
16	Sheath+ion energy 5 to 3; Bayesian+plasma diagnostics 7 across 2015–2025	[CALC]	sums of Table F.1 columns A and B

TEAM ASTROPHAGE

B Evidence Gaps and Next Steps

Every **[DATA GAP]** marked in the body is reproduced below.

#	Claim needed	Why unavailable	How to get it
1	The magnitude, in volts, of the documented Langmuir-versus-emissive probe disagreement inside the presheath	The abstract confirms the disagreement qualitatively and states it is not proportional to T_e, but the numerical magnitude sits behind the publisher's paywall	Institutional library access to Plasma Sources Science and Technology 29(2) 025015, or direct correspondence with the authors
2	The dollar cost of a single electric propulsion life-qualification campaign	No dollar figure appears in any of the NASA NEXT or HERMeS documents reviewed; the literature is uniformly qualitative on cost, using phrases such as "long and expensive lifetime tests"	Query USAspending.gov for the Aerojet Rocketdyne AEPS contract award value, which is the closest available public proxy; or request the figure directly from NASA Glenn
3	The cost of semiconductor fab downtime per hour, and the cost of plasma process excursions	Every per-hour figure encountered in research (ranging from $2 million to over $8 million per hour) traced back to search engine optimisation content farms and vendor marketing blogs with no primary citation. No figure was found in any SEMI publication, IEEE Transactions on Semiconductor Manufacturing paper, or named fab disclosure	A SEMI or SEMATECH publication, a peer-reviewed IEEE Transactions on Semiconductor Manufacturing paper, or a named fab's own public disclosure. This is the most commercially significant open gap in the dossier
4	The average selling price of a plasma etch tool	Lam Research, Applied Materials and Tokyo Electron do not publish per-tool prices; the figure circulating for one recent tool appears only in a paywalled analyst newsletter's search snippet and could not be read in the article itself	Analyst subscription (SemiAnalysis, TechInsights), or a fab equipment procurement record obtained under a freedom-of-information request
5	An enumerated list of the 46 member organisations by name	The count is now anchored to a dated archival snapshot and is reproducible [S33], but members were counted by distinct logo assets, which mis-counts any member listed without a logo or with more than one	Transcribe the member names from the archived snapshot into an appendix, converting a logo-asset count into a name list
6	Awarded federal contract values for every named incumbent diagnostics vendor, not only Impedans	Only one vendor's award history was queried; the equivalent query was not run for Hiden Analytical or the other named suppliers, so the displaced-spend cross-check rests on a single recipient	Re-run the same awards API query, filtered by recipient name, for each named incumbent, and report the full range rather than one vendor's
7	A worldwide census of electric propulsion and thruster vacuum test facilities	No authoritative count exists in any public source searched; named facilities were verified individually but no register or survey was found	A manual review of International Electric Propulsion Conference proceedings, which is the likeliest venue for such a survey, or a direct enquiry to the Electric Rocket Propulsion Society

TEAM ASTRA SPACE

#	Claim needed	Why unavailable	How to get it
8	A dated GBP/USD exchange rate, to permit direct comparison of the sterling licence band against dollar-denominated research awards	No exchange rate was retrieved or cited during this assessment, and using an undated rate would introduce an unsourced number into the report's central commercial comparison	Take a published central-bank or exchange reference rate for a stated date and restate both figures in one currency; note that against the largest on-topic award of $399,999.56 the break-even is approximately 0.52 USD per GBP **[CALC]**, so at every realistic rate the top of the licence band exceeds that award; the ordering is not rate-dependent, and the defensible statement is that the two are of the same order of magnitude
9	The identifiability map: the eigenvalue spectrum, condition number, Cramér–Rao comparison and resolvable/degenerate partition of the operating space	The team reports that the Fisher-information machinery exists but that none of these was computed, and that the n_0–T_e degeneracy ($\Gamma_i \propto n_0\sqrt{T_e}$) is argued analytically rather than measured **[C]**. The absence is stated by the team rather than concealed	Compute the spectrum and publish the partition of the operating space into resolvable and degenerate regions. This is now the single highest-value open item in the file: R1, identifiability, is the joint highest-scored risk in the register and is the one supported by the least evidence of any risk in it
10	Whether the applicant qualifies for the WIPO 90 percent fee reduction available to applicants from certain states	The reduction exists and is documented, but eligibility depends on a current list of qualifying states that was not checked in this research pass	Check the applicant's nationality and residence against the current WIPO schedule of fee reductions, which is published and free to consult
11	Whether plasma diagnostic inversion software for fusion and electric propulsion applications falls under any export control or dual-use classification	Assessed as plausible from the application domains, but no specific classification was checked against any control list in this research pass	A review against the relevant national export control list, or an opinion from a trade compliance adviser, before any international collaboration or code release
12	Any Indian national mission dedicated to fusion or plasma technology, with a published outlay	No such standalone mission was found on any Department of Atomic Energy, Department of Science and Technology, or Press Information Bureau page that could be opened; India's fusion effort is delivered institutionally through the Institute for Plasma Research rather than as a funded mission	Direct enquiry to the Department of Atomic Energy, or review of the DAE annual report, which may state programme-level allocations
13	Verified capability pages for PPPL, the University of Michigan Plasmadynamics and Electric Propulsion Laboratory, LPP at École Polytechnique, and the Air Force Research Laboratory	All four sites rejected automated access with HTTP 403 or a Cloudflare challenge; these are real and well-known groups but no official page could be opened to verify their current programmes	Manual browser retrieval of the four sites, or direct contact with the named laboratories
14	Any current funding, budget, runway, team size or institutional support figure for the project	The Context Block states no funding information, and no company or financial record exists to search	Direct disclosure from the applicant, which is a reasonable request at the first committee meeting

TEAM-STREPHACY

#	Claim needed	Why unavailable	How to get it

Total data gaps recorded: 14.

The gaps most worth closing before presenting

Assessment: Ranked by how much they would change a committee's decision:

1. **Any T2 accuracy figure at all.** This is the only gap that converts the entire dossier from a proposal into evidence, and it requires compute and time rather than money or hardware.
2. **A literature-based prior-art clearance.** The novelty risk is scored 16 and rests on dense non-patent art (§11.2). A keyword patent search does not address it.
3. **A written laboratory access agreement.** Facility access is scored 12, and AIC-Plasmatech at the Institute for Plasma Research is a specific, identified route [S86].
4. **A primary source for semiconductor process-excursion cost.** The entire semiconductor value argument currently rests on an unsourced number that this report refuses to quote.
5. **A worldwide census of electric propulsion test facilities.** It is the denominator of the largest single component of the serviceable market.

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C Patent Search Queries, Verbatim

The whitespace assessment in §11.2 rests on these twenty-three searches and on nothing else. They were run on 6 August 2026 over Google Patents and USPTO full-text search. They are reproduced exactly as submitted so that the absence claim can be reproduced or falsified.

Assessment: The disclosed limits of this set are as stated in §11.2: keyword-based and English-only, with no Cooperative Patent Classification sweep, no citation-network expansion, and no Japanese, Korean or Chinese language searching. Espacenet, PATENTSCOPE and Lens.org returned no usable result pages, so coverage is effectively Google Patents plus USPTO full text. **Assessment: A further limitation, and a material one:** several of the queries are long natural-language strings rather than short Boolean or classification-anchored expressions. Query sets of that shape return few or no results for almost any subject, so a null result from them carries correspondingly little information. The whitespace finding should be re-tested with short Boolean and CPC-anchored queries before it is relied upon.

#	Query string as submitted
Q1	`ion energy distribution plasma sheath determination measurement claim`
Q2	`optical emission spectroscopy plasma etch endpoint detection Verity Instruments`
Q3	`non-invasive plasma parameter determination optical emission spectroscopy model actinometry electron temperature`
Q4	`Bayesian inference plasma diagnostic semiconductor process control uncertainty quantification`
Q5	`laser induced fluorescence plasma ion velocity distribution measurement apparatus`
Q6	`Thomson scattering plasma diagnostic apparatus electron density temperature measurement`
Q7	`virtual metrology plasma etch chamber machine learning model based sensing predict wafer`
Q8	`plasma sheath model ion energy control tailored waveform RF bias Advanced Energy Industries`
Q9	`Bayesian inversion fusion multiple plasma diagnostics integrated data analysis forward model`
Q10	`Impedans Semion patent ion energy analyzer plasma sensor`
Q11	`Lam Research optical emission spectroscopy plasma chamber sensor machine learning virtual sensor`
Q12	`"Verity Instruments" patent assignee optical emission spectrometer plasma`
Q13	`neural network optical emission spectra plasma predict ion energy plasma parameter estimation inverse model`
Q14	`collisional radiative model optical emission spectroscopy determine plasma electron energy distribution function apparatus`
Q15	`university assignee plasma sheath optical diagnostic ion energy inference CEA Max Planck national laboratory`
Q16	`interferometry plasma electron density in situ measurement semiconductor processing chamber non-invasive`
Q17	`plasma optical emission Bayesian estimate plasma state uncertainty claim semiconductor`
Q18	`Applied Materials plasma sheath voltage determination optical measurement ion energy in situ sensor`
Q19	`Bayesian plasma process control semiconductor chamber posterior probability sensor patent granted`
Q20	`laser induced fluorescence plasma processing chamber measuring species density apparatus semiconductor`
Q21	`Espacenet ion energy distribution function optical emission spectroscopy inference plasma sheath model`

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#	Query string as submitted
Q22	`Bayesian inverse problem plasma diagnostic reconstruct plasma parameters uncertainty granted patent tokamak`
Q23	`optical emission sheath measure ion energy sheath optical sensor determine sheath voltage plasma processing control`

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D Revisions and Corrections

Corrections made to this assessment during preparation, recorded so that a reader comparing an earlier circulated copy against this one can see what changed and why. No entry below is an open issue; each was resolved in the text.

#	Item	Correction
1	Fusion funding attribution	The $14.24 bn total was initially described as private investment. The source splits it as $13.26 bn private and $0.98 bn public; the total overstated private capital by $0.98 bn. Corrected throughout [S25].
2	Year-on-year fusion growth	A 69% year-on-year increase was quoted. That percentage does not appear in the cited report and was withdrawn; the two annual totals are quoted instead [S25].
3	Validation cost envelope	The upper bound was drawn from NSF award 2117582 ($989,876), which funds an airborne coronal emission surveyor, a solar-astronomy instrument rather than a laboratory plasma diagnostic. It was removed and the envelope narrowed to the three on-topic whole-project awards [S110].
4	Anomalous-transport parameter	A tunable coefficient range of 0 to 1 was attributed to the cited review. The range does not appear in that source and was withdrawn; both codes are now described in the source's own terms [S2].
5	Serviceable market count	The organisation count was taken from a live membership page that does not reproduce week to week. It is now counted from a dated archival snapshot and cited to it [S33].
6	Exit-pattern claim	The report stated that every acquisition in this domain bought hardware with software attached. Tech-X, standalone simulation software, is a counter-example within the same table. Restated as two patterns differing by an order of magnitude [S64].
7	Readiness level	The assessment initially recorded TRL 2. Testing all four clauses of the software criterion showed two evidence clauses unmet, and the assignment was lowered to TRL 1 [S57].
8	Readiness level, superseded	The team subsequently reported the closed-loop synthetic benchmark executed, which satisfies both outstanding clauses. The assignment was raised to **TRL 2** [C]. The benchmark's results were not at that point supplied to this assessment and are not reproduced in it.
9	Benchmark results incorporated	The team supplied the benchmark output. T0/T1/T2, coverage and the negative four-channel finding were added (§8.2); maturity scores for Technology, Validation and Software were raised from 2/1/2 to 3/3/3; risk R2 was lowered from 15 to 10; the subject technology was plotted on the accuracy axis of Figure 7.1 for the first time. The identifiability map remains uncomputed and is now the leading open item.

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F Underlying Data Tables

Every figure in this report is generated by code from the tables below. Every *plotted* mark traces to a value in these tables. **Assessment:** Some figures also carry explanatory annotations (a licence band, a seat multiplier, a compute projection) whose values are stated and tagged in the body rather than repeated here; the tables cover the data, not the marginalia.

F.1 Bibliometric counts (Figure 16.1)

Database: OpenAlex, articles and conference papers only, accessed 6 August 2026 [S109]. 2026 is a partial year.

Table F.1: Publication counts by year and query series.

Year	A	B	C	D	E	F2	G	H
2015	5	0	59	2	87	3	13	5
2016	5	0	77	3	133	3	16	5
2017	5	0	54	4	124	6	15	6
2018	1	0	70	4	118	7	10	7
2019	0	0	56	7	123	1	9	5
2020	3	0	68	7	110	6	12	7
2021	3	1	72	5	128	6	15	13
2022	3	0	58	2	92	9	9	12
2023	1	1	64	3	97	6	25	8
2024	2	1	83	4	137	10	12	22
2025	3	4	89	6	156	19	29	43
2026*	2	0	61	4	122	8	23	33

*Partial year, excluded from all fits. A = “plasma sheath” AND “ion energy”; B = “Bayesian inference” AND “plasma diagnostics”; C = “laser induced fluorescence” AND plasma; D = “Thomson scattering” AND “plasma diagnostic”; E = “Hall thruster”; F2 = “Bayesian” AND “tokamak”; G = “inverse problem” AND plasma; H = “uncertainty quantification” AND plasma.

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F.2 Competitive matrix coordinates (Figure 7.1)

Table F.2: Competitive positioning coordinates. A blank accuracy cell means the value is not published, and the entity is plotted in the axis gutter rather than in the plane.

Entity	Channels	Error %	Error basis	Source
Minerva at JET†	10	n/a	n/a	[S18]
Minerva at W7-X	4	n/a	n/a	[S5]
IDA at ASDEX Upgrade†	4	n/a	n/a	[S4]
D-MIBAS at MAST-U	3	5	n_e vs SOLPS-ITER synthetic truth	[S6]
Hall thruster Bayesian calibration	n/a	10	held-out flow rates and pressures	[S20]
Bayesian IVDF from LIF	1	n/a	n/a	[S48]
Impedans Quantum	2	n/a	n/a	[S49]
Impedans Semion	1	n/a	n/a	[S40]
Impedans Semion 3keV	1	n/a	n/a	[S40]
Hiden EQP	1	n/a	n/a	[S41]
Scientific Systems SmartProbe	1	n/a	n/a	[S50]
Plasma Sensors VGPS	1	n/a	n/a	[S51]

Entity	Channels	Error %	Error basis	Source
Impedans Langmuir	1	n/a	n/a	[S40]
Subject technology	4	3.99	**T2 honest/mismatched; synthetic; self-reported**	**[C]**

†Channel count taken from the abstract or citation record only: the publisher pages for these two entries returned HTTP 402 and HTTP 403 respectively to automated retrieval and could not be read in full. The Hiden entry below is likewise recorded from indexed page text rather than from the page itself. These three coordinates are the least well-evidenced in the figure.

F.3 Market sizing inputs (Figure 5.1)

Table F.3: Market sizing underlying values.

Item	Value	Source
Adjacent context, not addressable		
Wafer fab equipment, all types, 2025	$115.7 bn	[S24]
Semiconductor equipment total, 2025	$133.0 bn	[S24]
Fusion industry funding, total to date ($13.26 bn private, $0.98 bn public)	$14.24 bn	[S25]
Space propulsion market, 2024	$10.21 bn	[S26]
Fusion industry funding, 12 months to July 2026	$4.48 bn	[S25]
Satellite propulsion market, 2024	$2.60 bn	[S27]
Counted organisations		
Private fusion companies	46	[S33]
Fusion education and research partners	21	[S33]
Plasma etch tool OEMs	10	named vendor list
Named EP test facilities (a floor)	7	facility pages
Total	**84**	**[CALC]**
Price anchors		
COMSOL seat licence	£3,025 / seat / year	[S32]
Licence value at 1 seat each	£254,100 / year	**[CALC]**
Licence value at 2 seats each	£508,200 / year	**[CALC]**
Licence value at 3 seats each	£762,300 / year	**[CALC]**
SBIR Phase II ceiling (reference value, *not* plotted, see §15.2)	$2,153,927	[S74]

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F.4 Maturity scores (Figure 9.1, Panel A)

Table F.4: Technology maturity scores and their justifications. All scores are the analyst's assessment.

Axis	Score	Justification
Technology	3	Raised from 2. All four NASA software TRL 2 clauses reported met and the benchmark executed with results supplied **[C]**. Held at 3: every measurement is synthetic, and no laboratory data exists.
Validation	3	Raised from 1. T0/T1/T2 executed and reported, with 8 of 10 coverage at the 90 % level **[C]**. Held at 3: coverage is marginal on $N = 10$, no identifiability map was computed, and nothing has met real data.
Software	3	Raised from 2. Four-layer chain implemented and run end to end, with 2733 passing tests and 98 % core coverage reported **[C]**. Held at 3: the licensing decision is open, so no result is externally reproducible.
Documentation	4	Baseline v1.0 with explicit gate structure and instrument realism to commercial part numbers; materially ahead of every other axis.

Axis	Score	Justification
Commercialization	1	No company, no revenue, no pricing, no customer engagement, and no analyst category.
IP	1	No patent filed; thin patent landscape reflects dense non-patent prior art, which is a novelty risk rather than whitespace.
Partnerships	1	No partnership exists; targets identified but no relationship claimed.

F.5 Risk scores (Figure 9.1, Panel B)

Table F.5: Risk register coordinates.

ID	Category	P	I	P×I	Risk
R1	Technical	4	4	16	Identifiability limits across operating space
R2	Validation	3	5	15	T2 accuracy too wide to be decision-useful
R3	Technical	4	2	8	Compute budget wrong (but gated)
R4	IP	4	4	16	Novelty defeated by non-patent prior art
R5	Competition	3	4	12	D-MIBAS closes its gaps first
R6	Competition	3	3	9	Incumbent absorbs the capability
R7	Adoption	3	3	9	No procurement precedent for calibrated uncertainty
R8	Market	5	3	15	Licence revenue alone insufficient
R9	Execution	3	4	12	No access to a validation facility
R10	Regulatory	2	3	6	Export control (evidence base weaker than others)

F.6 Roadmap stages (Figure 17.1)

Table F.6: Roadmap stages and exit gates. No durations are given because no schedule has been validated.

#	Stage	Content	Exit gate
0	Baseline v1.0 specification	Architecture, forward/inverse model, validation method documented; nothing run	TRL 1 met in full; TRL 2 met on design criteria only
1	Measure the compute assumption *(part done)*	Gate G-1.4: measure GPU particle-push throughput, correct the cost model	Projected runtime replaced by a measured one
2	Execute the T0/T1/T2 benchmark	Closed-loop synthetic validation with deliberate model mismatch	TRL 2 complete (NASA software: experiments performed with synthetic data)
3	Publish the identifiability map	State unresolvable regions and by how much; submit to peer review	Independent scrutiny of the honest tier
4	Laboratory validation	Real optical data at a partner facility against a reference intrusive measurement	TRL 3 reached (validation against partially representative data)

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